

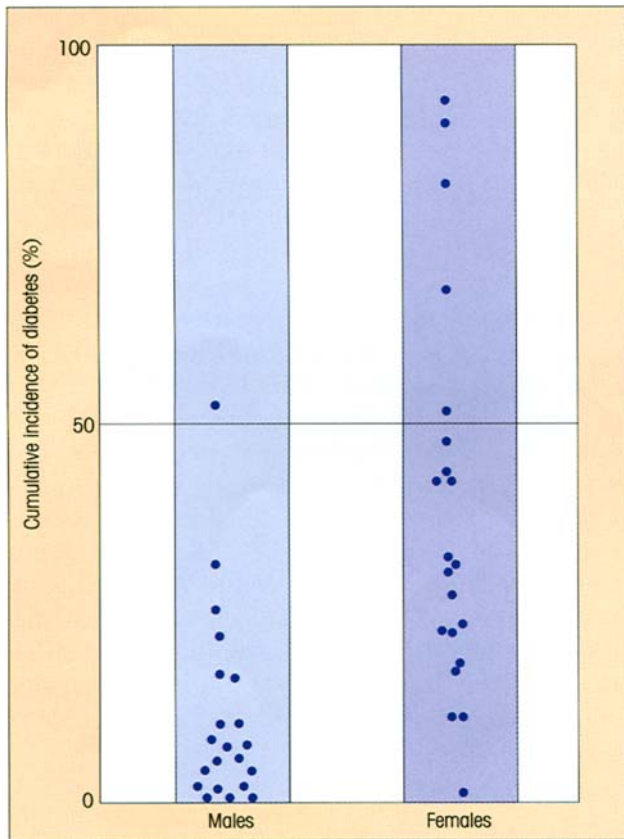


Figure 19.8. The incidence of spontaneous diabetes in geographically dispersed colonies of NOD mice at 20 weeks of age. Each point represents a single colony. The extreme spread of values is not attributable to genetic drift to any significant extent. The lower incidence in males is particularly evident. (Data adapted from Pozzilli P., Signore A., Williams A.J.K. & Beales P.E. (1993) NOD mouse colonies around the world. *Immunology Today* 14, 193.)

germ-line antibodies, mostly IgM, although a proportion belong to the IgG and IgA classes. They include a basic set of autoantibodies with low affinity reactivity for multiple specificities and cross-reactivity with common bacterial antigens usually of a carbohydrate nature. One can see this as a strategy which ensures that preliminary excitation of cells by these internal network interactions will provide bacterial protection, especially since the polymeric nature of the carbohydrate antigens ensures that the IgM antibodies, even though of low affinity, can bind with high avidity to the microbes.

Other functions for these natural antibodies have been proposed which are not mutually exclusive: Grabar viewed them as transporting agents responsible for scavenging effete body components. Others envisage a homeostatic role in which they actually prevent stimulation of autoreactive cells in the conventional B-2 cell population either by masking autoanti-

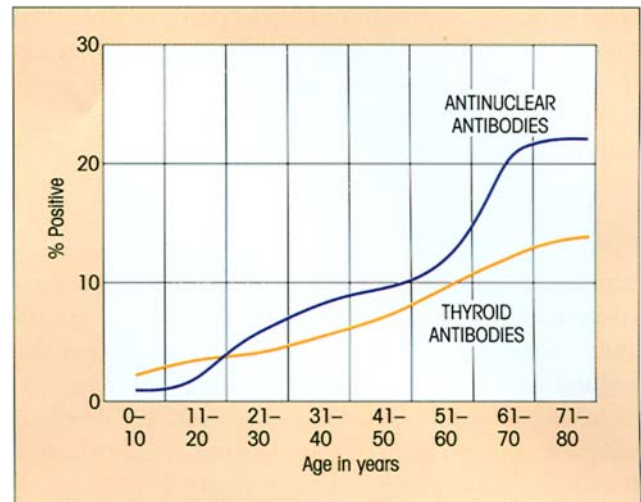


Figure 19.9. Incidence of autoantibodies in the general population. A serum was considered positive for thyroid antibodies if it reacted at a dilution of 1:10 in the tanned red cell test or neat in the immunofluorescent test, and positive for antinuclear antibodies if it reacted at a dilution of 1:4 by immunofluorescence. The class of antibody was not determined in these studies.

gen epitopes or by overall idiotypic regulation. The latter view receives some encouragement from a report that the IgM fraction of normal serum can block binding of autologous IgG F(ab')₂ fragments to a range of autoantigens. These natural autoantibodies presumably make a contribution to the so-called 'nonspecific' Ig binding of otherwise normal sera to test autoantigens, but slightly stronger reactions are evident in a small proportion of younger people and the incidence increases with age (figure 19.9). Although not associated with clinically apparent tissue damage, it should be noted that, in the case of the thyroid and stomach at least, biopsy has linked the presence of raised titers of antibody, especially of the IgG class, with minor thyroiditis or gastritis lesions (as the case may be), and postmortem examination has identified 10% of clinically normal middle-aged women with significant but limited degrees of lymphadenoid change in the thyroid, similar in essence to that characteristic of Hashimoto's disease.

As enthusiasts for symmetry and order might have predicted, there appears to be an analogous T-cell population, of phenotype CD3⁺ CD4⁺ CD8⁻ bearing the B-cell marker B220, containing large internally activated cells which react strongly with self-T-cells and are expanded in early life. It is possible that they connect to the CD5⁺ B-cell network through idiotypic interactions with cells of this lineage present in the thymus. The reader may be surprised to learn that, in generating T-cell lines, it is not an uncommon experience to isolate

cells which proliferate and release IL-2 in response to autologous class II-positive feeder cells; even allowing for the fact that the presence of these feeders in the cultures will tend to select for such autoreactive cells, it would not have been predicted that cells with these specificities would be permitted to roam around freely in the body unless constrained in some way. This brings us back to the ideas of the 'immunological homunculus' in which dominant autoantigens in the body are imprinted on the immune system and the T-cells which recognize them are heavily controlled by regulatory T-cells (cf. p. 207).

Although the effector cells of autoimmune disease are present in normal individuals, abnormal conditions must be required for their stimulation. In experimental models of organ-specific disease, such as that induced in the thyroid by injection of thyroglobulin in complete Freund's adjuvant, the effector T-cells and the plasma cells making high affinity IgG autoantibodies are generated in normal animals. Complete Freund's will not produce antibodies to double-stranded DNA, Sm or other autoantigens typical of nonorgan-specific disorders and this may be telling us that the relevant antigen-specific helper T-cells are not available in the normal repertoire. However, if T-cells are stimulated by radically different approaches, nonorgan-specific antibodies can be coaxed out of normal animals; in one system, allogeneic T-cells inducing a graft-vs-host (g.v.h.) reaction are stimulated by, and thence polyclonally activate, class II-bearing B-cells, while another involves immunization with a public anti-DNA idiotype (16/6) in complete Freund's.

IS AUTOIMMUNITY DRIVEN BY ANTIGEN?

This is not such a silly question as it might appear since lymphocytes can be stimulated not only by antigens but also by anti-idiotypes and by superantigens as well as other polyclonal activators. And if the answer is in the affirmative, is the self-molecule an autoimmunogen or just an autoantigen, i.e. does it drive the autoimmune response or is it merely recognized by its products?

Organ-specific disease

First, some direct evidence straight from the shoulder. The Obese strain (OS) chicken spontaneously develops precipitating IgG autoantibodies to thyroglobulin and a chronic inflammatory antithyroid response which destroys the gland so causing hypothyroidism. If the source of antigen is removed by neonatal

thyroidectomy, no autoantibodies are formed. Injection of these animals with normal thyroglobulin then induces antibodies. Thyroidectomy of chickens with established thyroiditis is followed by a dramatic fall in antibody titer. Conclusions: the spontaneous antithyroglobulin immunity is initiated and maintained by autoantigen from the thyroid gland. Furthermore, since the response is completely T-cell dependent, we can infer that both B- and T-cells are driven by thyroglobulin in this model. An entirely parallel study in NOD mice showed that destruction of the β -cells in the pancreas by alloxan switched off the stimulus to autoantibody production.

As usual, human disease is a tougher nut to crack and one has to rely on more indirect evidence. T-cell lines have been established from thyrotoxic glands and it has been possible to show direct stimulation by whole thyroid cells. The production of high affinity IgG autoantibodies accompanied by somatic mutation is taken as powerful evidence for the selection of B-cells by antigen in a T-dependent response. The reason for this, simply, is that high affinity IgG antibodies only arise through mutation and selection by antigen within germinal centers (cf. p. 190). Suffice it to say that ample evidence for somatic mutation and high affinity antibodies has been reported. More indirect is the argument that, when antibodies are regularly formed against a cluster of epitopes on a single molecule (e.g. thyroglobulin) or of antigens within a single organ (e.g. thyroglobulin plus thyroid peroxidase), it is difficult to propose a hypothesis which does not depend finally on stimulation by antigen.

Systemic autoimmunity

The question is even harder to answer here, particularly since antigen removal is impossible. With respect to B-cells, the same arguments marshaled for organ-specific disease obtain, i.e. high affinity mutated IgG autoantibodies directed often to antigen clusters such as the constituents of the nucleosome. (Readers who like delving into mechanisms should consult figure 19.12b.5 and then figure 19.13c to follow the manner in which an activated B-cell, specific for one component in a complex, can present epitopes on a second constituent of the same complex to an activated T-helper.)

T-cells are critical for such responses and, indeed, depletion of CD4 T-cells in NZB or NZB $\times$ W mice abrogates autoantibody production. Fine so far, but from there on we are in black box territory since we are woefully ignorant of the antigen specificity of the T-cells. So much so that more radical hypotheses are tendered. One of these postulates that we are really dealing with

hypersensitivity responses to microorganisms which are difficult to identify (see above), although one still has to account for the autoantibodies produced and their known pathogenic role in certain diseases (e.g. complexes in SLE). Another view which has been seriously mooted is that the T-cells do not see conventional antigen at all, clearly the case with DNA responses, but instead are devoted to the recognition of idiootype; SLE for example would be an '**idiotype disease**' resulting from network breakdown. Conceivably, the network may break down spontaneously or be 'hacked' into by microbes (cf. figure 19.14). We may notice that immunization of mice with monoclonal autoantibodies to DNA, ribonucleoprotein (RNP) and Sm has stimulated production of the corresponding antibody bearing the original idiootype; this must have involved an internal network.

Is autoantigen available to the lymphocytes?

Our earliest view, with respect to organ-specific antibodies at least, was that the antigens were sequestered within the organ, and through lack of contact with the lymphoreticular system failed to establish immunological tolerance. Any mishap which caused a release of the antigen would then provide an opportunity for autoantibody formation. For a few body constituents this holds true, and in the case of sperm, lens and heart for example, release of certain components directly into the circulation can provoke autoantibodies. But, in general, the experience has been that injection of *unmodified* extracts of those tissues concerned in the organ-specific autoimmune disorders does not readily elicit antibody formation. Indeed, detailed investigation of the thyroid autoantigen, thyroglobulin, has disclosed that it is not completely sequestered within the gland but gains access to the extracellular fluid around the follicles and reaches the circulation via the thyroid lymphatics (figure 19.10). Even in the brain, the ability of systemically injected T-cell clones specific for myelin basic protein to induce encephalitis (cf. p. 435) reveals the exposure of the target antigen. In fact, in the majority of cases—e.g. red cells in autoimmune hemolytic anemia, RNP and nucleosome components present as blebs on the surface of apoptotic cells in SLE, and surface receptors in many cases of organ-specific autoimmunity—the autoantigens are readily accessible to circulating lymphocytes.

Presumably, antigens present at adequate concentrations in the extracellular fluid will be processed by professional APCs, but for autoantigens associated with cells, the derivative peptides will only interact 'meaningfully' with specific T-cells if there are ap-

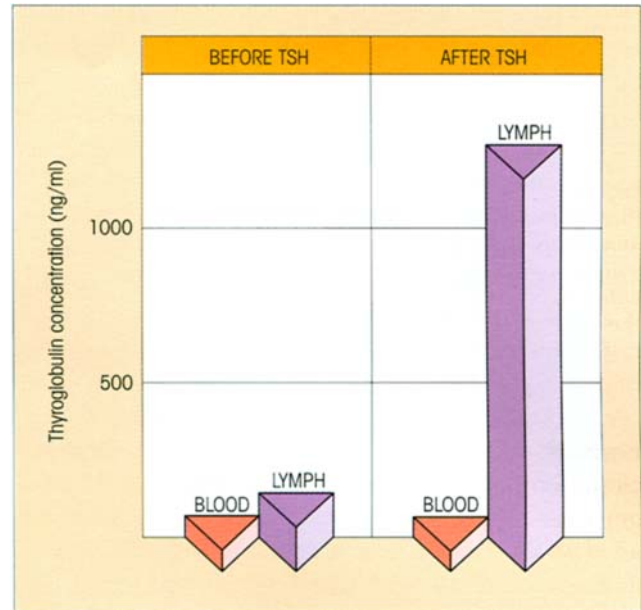


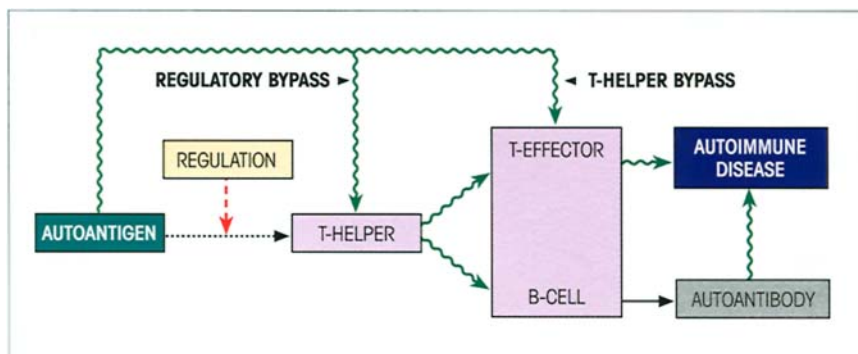
Figure 19.10. Thyroglobulin in the cervical lymph draining the thyroid in the rat. The concentration of thyroglobulin is increased after injection of pituitary thyroid-stimulating hormone (TSH), showing that the release from thyroid follicles is linked to the physiological activity of the acinar cells. Colloid is taken up at the apical margin and thyroglobulin cleaved proteolytically to thyroid hormones which are released together with undegraded protein from the base of the cell. (From Daniel P.N., Pratt D.E., Roitt I.M. & Torrigiani G. (1967) *Quarterly Journal of Experimental Physiology* 52, 184.)

propriate MHC surface molecules, if the concentration of processed peptide associated with them is significant and, for resting T-cells, if costimulatory signals can be given. As we shall see, these are important constraints.

CONTROL OF THE T-HELPER CELL IS PIVOTAL

The message then is that we are all sitting on a mine-field of self-reactive cells, with potential access to their respective autoantigens, but since autoimmune disease is more the exception than the rule, the body has homeostatic mechanisms to prevent them being triggered under normal circumstances. Accepting its limitations, figure 19.11 provides a framework for us to examine ways in which these mechanisms may be circumvented to allow autoimmunity to develop. It is assumed that the key to the system is control of the autoreactive T-helper cell since the evidence heavily favors the T-dependence of virtually all autoimmune responses; thus, interaction between the T-cell and MHC-associated peptide becomes the core consideration. We start with the assumption that these cells are

Figure 19.11. Autoimmunity arises through bypass of the control of autoreactivity. The constraints on the stimulation of self-reactive helper T-cells by autoantigen can be circumvented either through bypassing the helper cell or by disturbance of the regulatory mechanisms.



normally unresponsive because of clonal deletion, clonal anergy, T-suppression or inadequate autoantigen presentation. Immediately, one could conceive of an *abnormal* degree of responsiveness to self-antigens as a result of relatively low intrathymic expression of a particular molecule (cf. p. 405). Abnormalities in the signaling pathways affecting the thresholds for positive and negative selection in the thymus would also affect subsequent responsiveness to peripheral autoantigens. So might defects in apoptotic cell death. It would be interesting to know whether the innate resistance of the NZB mouse to tolerization by a protein antigen such as bovine serum albumin can be nailed to one of these causes or whether it is a consequence of defects in regulatory cells (see below).

AUTOIMMUNITY CAN ARISE THROUGH BYPASS OF T-HELPERS

Provision of new carrier determinant

Allison and Weigle argued independently that, if autoreactive T-cells are tolerized and thereby unable to collaborate with B-cells to generate autoantibodies (figures 19.12a and 19.15), provision of new carrier determinants to which no self-tolerance had been established would bypass this mechanism and lead to autoantibody production (figures 19.12b and 19.15).

1 Modification of the autoantigen

A new carrier could arise through some modification to the molecule, for example by defects in synthesis or by an abnormality in lysosomal processing yielding a split product exposing some new groupings (figure 19.12b.1). Aside from the possibility of post-translational modifications, such as subtle changes in glycosylation patterns as seen in the low galactosylation of the Fc γ sugar chains in rheumatoid arthritis, many studies on spontaneous autoimmune disease

have failed to reveal an abnormality in the antigen. Remember the experiment in which neonatal thyroidec-tomized Obese strain chickens make autoantibodies if injected with thyroglobulin prepared from *normal* chickens, suggesting that the immunological response rather than the antigen is abnormal. Nonetheless, there may be defects in the iodine metabolism of the gland itself in this strain (figure 19.4), and recent work has shown that the severity of thyroiditis is ameliorated when the birds are put on a low iodine diet.

Modification can also be achieved through combination with a drug (figure 19.12b.3). The autoimmune hemolytic anemia associated with the administration of α -methyl dopa might be attributable to modification of the red cell surface in such a way as to provide a carrier for stimulating B-cells which recognize the rhesus antigen. This is normally regarded as a 'weak' antigen and would be less likely to induce B-cell tolerance than the 'stronger' antigens present on the erythrocyte. Isoniazid may produce arthritis associated with nuclear antibodies and, unlike most other cases of drug-induced autoimmunity, synthesis of these antibodies is said to continue after cessation of drug therapy. A high proportion of patients on continued treatment with procainamide develop nuclear antibodies and 40% present with clinical signs of SLE. Myasthenia gravis and symptoms of pemphigus have been described in some patients on penicillamine. It is not clear in every case whether the drug provides carrier help through direct modification of the autoantigen or of some independent molecule concerned in associative recognition.

2 Cross-reactions with B-cell epitopes

Many examples are known in which potential human autoantigenic B-cell epitopes are present on a microbial exogenous cross-reacting antigen which provides the new carrier that provokes autoantibody formation (figure 19.12b.2). The mechanism is spelt out in more

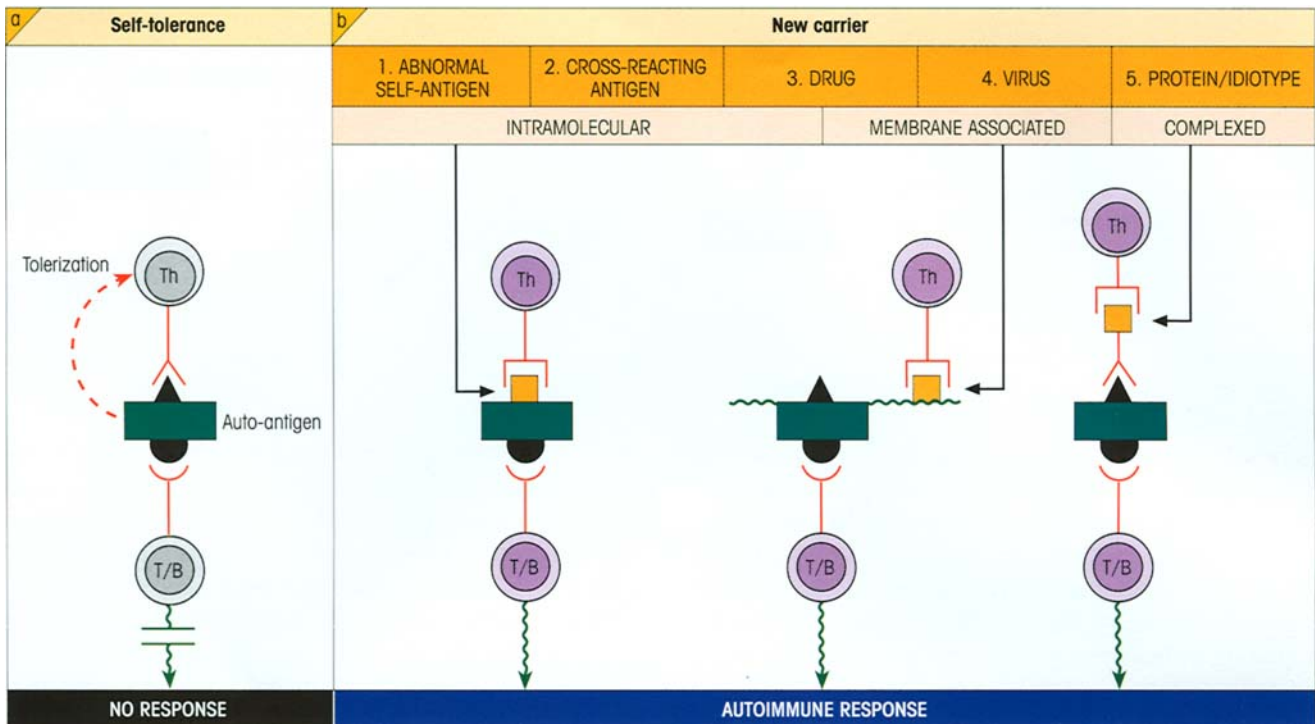


Figure 19.12. T-helper bypass through new carrier epitope (■) generates autoimmunity. For simplicity, processing for MHC association has been omitted from the diagram, but is elaborated in figure 19.13. (a) The pivotal autoreactive T-helper is unresponsive either through tolerance or inability to see a cryptic epitope. (b) Different mechanisms providing a new carrier epitope.

detail in figure 19.13a. Two low molecular weight envelope proteins of *Yersinia enterocolitica* share epitopes with the extracellular domain of the human thyroid-stimulating hormone (TSH) receptor; in rheumatic fever, antibodies produced to the *Streptococcus* also react with heart, and the sera of 50% of children with the disease who develop Sydenham's chorea give neuronal immunofluorescent staining which can be absorbed out with streptococcal membranes. Colon antibodies present in ulcerative colitis have been found to cross-react with *Escherichia coli* 014. There is also some evidence for the view that antigens common to *Trypanosoma cruzi* and cardiac muscle and peripheral nervous system provoke some of the immunopathological lesions seen in Chagas' disease.

3 Molecular mimicry of T-cell epitopes

The drawback with the Allison–Weigle model of cross-reaction of B-cell epitopes and the provision of a new T-cell carrier is that, once the cross-reacting agent is

eliminated from the body, and with it the T-cell epitope, the only way that the autoimmunity can be sustained is for the activated B-cell to capture circulating autoantigen and, after processing, present it to the T-helper (figure 19.13c). This is not possible for **cell-associated antigens** but their special link with T-cell recognition puts them in a totally different ballpark. In this case, if an infecting agent mimics an autoantigen by producing a **cross-reacting T-cell epitope**, the resulting T-cell autoimmunity could theoretically persist even after elimination of the infection. The autoantigen will normally be presented to the resting autoreactive T-cell as a **cryptic epitope** and by definition will be unable to provide an activating signal. The cross-reacting infectious agent will provide abundant antigen on professional APCs which can prime the T-cell and upregulate its adhesion molecules so that it now has the **avidity** to bind to and be persistently activated by the cryptic self-epitope presented on the target tissue cell provided that it is associated with the appropriate MHC molecule (figure 19.13b). Remember the transgenic cytotoxic T-cells (Tc) which could only destroy the pancreatic β -cells bearing a viral transgene when they were **primed** by a real viral infection (cf. figure 12.10). Recall also the tumor cells that could only be recognized by primed not resting T-cells (cf. figure 18.9). Theoretically, the resting T-cell could also be primed in a nonantigen-specific manner by a microbial **superantigen**.

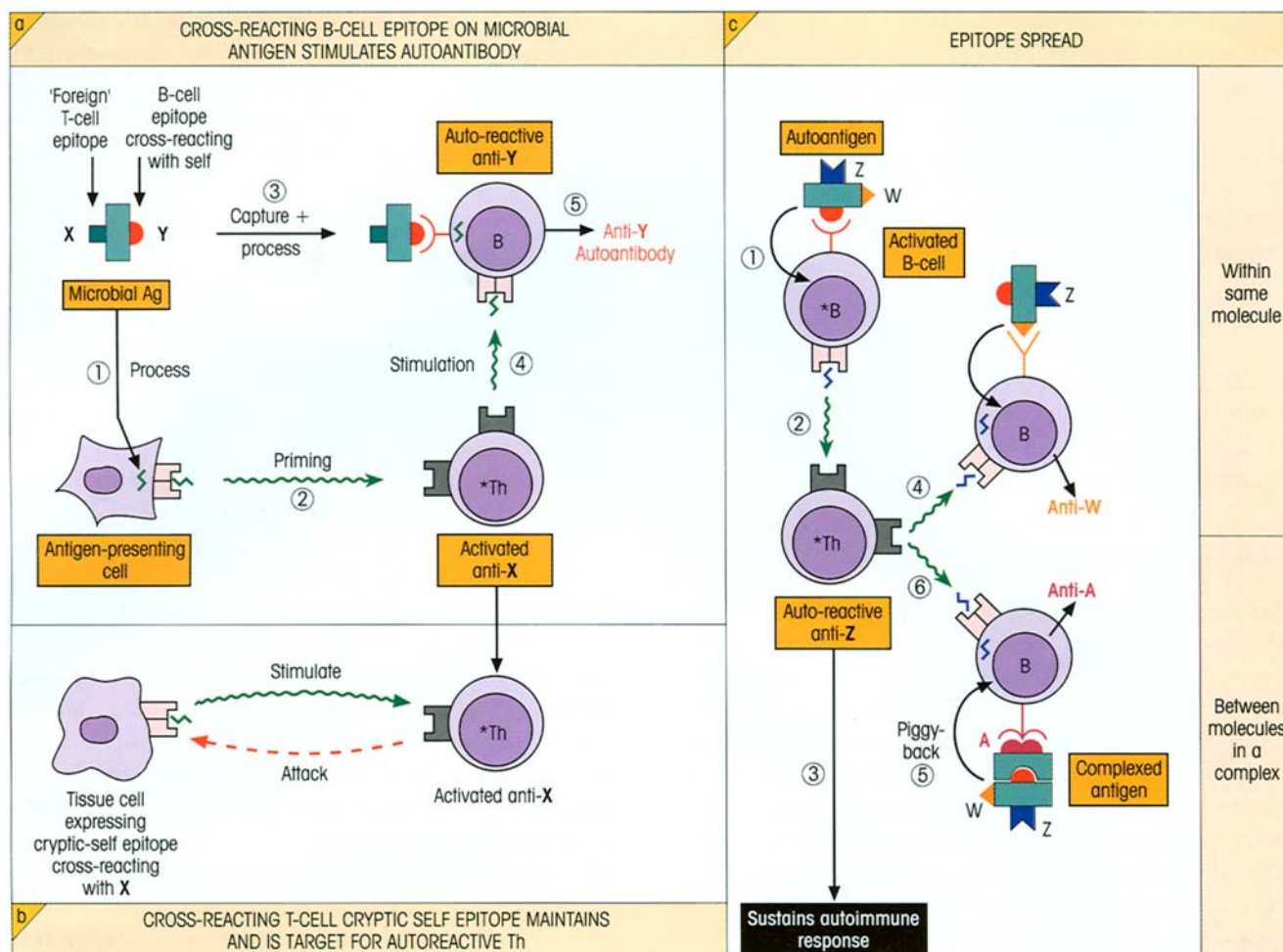


Figure 19.13. Mechanisms of microbial induction of autoimmunity and epitope spread. (This is a complex mouthful but digestion is recommended because these ideas are crucial. The more faint-hearted may require an ice-pack and persistence, but following the numbers should help.) (a) A microbial antigen bearing an epitope Y which cross-reacts with self and a foreign T-cell epitope X is (1) processed by an antigen-presenting cell, (2) activates the T-helper which (3) recognizes the processed X after capture by an anti-Y B-cell and (4) stimulates the B-cell to secrete anti-Y autoantibody. (b) The *activated* anti-X T-helper, as distinct from the *resting* cell, may recognize and be stimulated by a cross-reacting cryptic T-cell epitope expressed by a tissue cell. This will maintain the autoimmune response even after elimination of the microbe, because of the persistence of the self-epitope. The tissue expressing the epitope will also be a target for immunological attack. Note also that a T-helper primed nonspecifically by a polyclonal super-

antigen activator could also fulfil the same function of responding to a cryptic epitope. (c) If the autoantigen is soluble or capable of uptake and processing after capture by the activated autoreactive B-cell (1) (either from (a) or through nonspecific polyclonal activation), a new epitope can be presented on the B-cell class II which now stimulates an autoreactive (anti-Z) T-helper (2) which can now sustain an autoimmune response entirely through autoantigen stimulation (3). It can also produce epitope spread within the same molecule through helping a B-cell which captures the autoantigen through a new epitope W (4). It can also permit epitope spread to another component in an intermolecular complex such as nucleosomal histone-DNA or idiotype-positive (Id⁺) anti-DNA-DNA which is 'piggy-backed' into the B-cell (5) which presents processed antigen to the T-helper (6) in the cases cited, specific for histone or Id, respectively. *Denotes activation.

Although we have ascribed the dominant role of MHC alleles as risk factors for autoimmune diseases to their ability to present key antigenic epitopes to autoreactive T-cells, they might also operate in a quite distinct way. We may recollect that, during intrathymic ontogeny, T-cells are positively selected by weak interaction with self-peptides complexed with MHC. Now since around 50% of the class II peptides are MHC de-

rived (figure 5.22), then the mature T-cells leaving the thymus will have been selected with a strong bias to weak recognition of self-MHC peptides presented by class II. There must therefore be a major pool of self-reactive T-cells vulnerable to stimulation by exogenously derived cross-reacting epitopes which mimic these MHC peptides. Just so. The critical sequence QKRAAVDTY of the rheumatoid arthritis susceptibil-

ity allele HLA-DRB1*0401 (table 17.2) is closely similar to the QKRAAYDQY of the dnaJ heat-shock protein of *E. coli*, and this peptide presented by DQ causes proliferation of synovial T-cells from RA patients. In fact, a large number of microbial peptide sequences with varying degrees of homology with human proteins have been identified (table 19.5), although it should be emphasized at this stage that they only provide clues for further study. The mere existence of a homology is no certainty that infection with that organism will necessarily lead to autoimmunity because everything depends on several contingencies, including the manner in which the proteins are processed by the APCs, and we cannot predict, as yet, which peptides will be presented and in what concentration.

4 'Piggy-back' T-cell epitopes and epitope spread

One membrane component may provide help for the immune response to another (associative recognition). In the context of autoimmunity, a new helper determinant may arise through drug modification as mentioned above, or through the insertion of viral antigen into the membrane of an infected cell (cf. figure 19.12b.4). That this can promote a reaction to a pre-existing cell component is clear from the studies in which infection of a tumor with influenza virus elicited resistance to uninfected tumor cells. The appearance of cold agglutinins often with blood group I specificity after *Mycoplasma pneumoniae* infection could have a similar explanation. In a comparable fashion, T-cell help can be provided for a molecule such as DNA, which cannot itself form a T-cell epitope, by complexing with a T-dependent carrier, in this example a histone, or an anti-DNA idiotype to which T-cells were sensitized. For this mechanism to work, the helper component must still be physically attached to the fragment bearing the B-

cell epitope. When this is recognized by the B-cell receptor, the helper component will be 'piggy-backed' into the B-cell, processed and presented as an epitope for recognition by T-cells (figure 19.13c). By the same token, the autoimmune response can spread to other epitopes on the same molecule.

Idiotype bypass mechanisms

We have argued the evidence for internal regulated idiotype networks involving self-reactivity at some length. This raises the possibility of involving autoreactive lymphocytes with responses to exogenous agents through idiotype network connections, particularly since some autoimmune diseases are characterized by major cross-reactive idiotypes.

Thus, knowing that T-helpers with specificity for the idiotype on a lymphocyte receptor can be instrumental in the stimulation of that cell, it is conceivable that an environmental agent such as a parasite or virus, which triggered antibody carrying a public idiotype (cross-reactive idiotype, CRI), which happened to be shared with the receptor of an autoreactive T- or B-cell, could provoke an autoimmune response (figure 19.14b). Similarly, if it is correct that the germ-line idiotypes on autoantibodies generate a whole range of anti-idiotypes which mediate the response to exogenous antigens, then, by the same token, it is conceivable that antibodies produced in response to an infection may react with the corresponding idiotype on the autoreactive lymphocyte (figure 19.14a). For example, a hybridoma from a myasthenia gravis patient secreted an anti-Id to an acetylcholine receptor autoantibody; this anti-Id was found to react with the bacterial product 1,3-dextran. Finally, it is possible for Id network interactions to allow a viral infection to give rise to autoantibodies reacting with the viral receptor (figure 19.14c). Since viruses all bind to specific complementary receptors on the cells they infect, this sequence of events may have serious consequences; we note for example that β -adrenergic receptors are the surface targets for certain reoviruses and that rabies virus binds to the acetylcholine receptor.

Polyclonal activation

Microbes often display adjuvant properties through their possession of polyclonal lymphocyte activators such as bacterial endotoxins, which act by providing nonspecific inductive signals that bypass the need for specific T-cell help, either by stimulation of CD8 T-cells through upregulation of dendritic cell CD40 or by direct interaction with B-cell mitogen receptors (cf.

Table 19.5. Molecular mimicry: homologies between microbes and body components as potential cross-reacting T-cell epitopes.

Microbial molecule	Body component
Bacteria: Arthritogenic <i>Shigella flexneri</i> <i>Klebsiella</i> nitroreductase <i>Proteus mirabilis</i> urease <i>Mycobact. tuberculosis</i> 65 kDa hsp <i>E.coli</i> DNAJ hsp	HLA-B27 HLA-B27 HLA-DR4 Joint (adjuvant arthritis) RA shared DRB1 T-cell epitope
Viruses: Coxsackie B Coxsackie B EBV gp110 HBV octamer HSV glycoprotein Measles hemagglutinin Retroviral gag p32	Myocardium Glutamic acid decarboxylase RA shared DRB1 T-cell epitope Myelin basic protein Acetylcholine receptor T-cell subset U-1 RNA

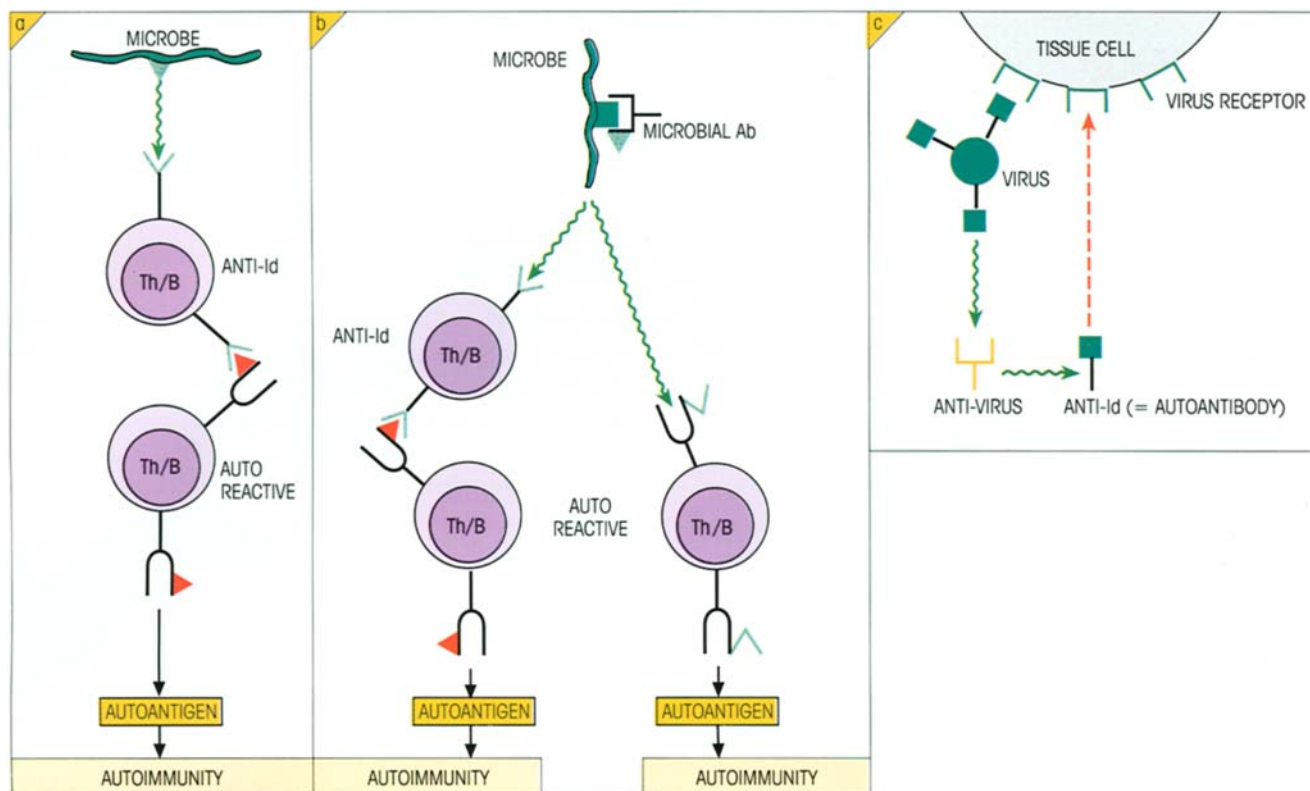


Figure 19.14. Idiotypic mechanisms leading to autoimmunity. (a) Microbial antigen cross-reacts with autoreactive lymphocyte Id. (b) Microbial antibodies either share Ids with or are anti-Id to autoreactive lymphocytes. (c) Anti-virus generates anti-Id which is autoantibody to viral receptor (Plotz).

characteristic of the different autoimmune disorders without the operation of some antigen-directing factor. We have already hinted at scenarios in which polyclonally activated B- or T-cells might contribute to a sustained autoimmune response (see legend to figure 19.13b and c).

p. 171). This can occur by direct interaction with the B-lymphocyte or indirectly through stimulating the secretion of nonspecific factors from T-cells or macrophages. The variety of autoantibodies detected in cases with infectious mononucleosis must surely be attributable to the polyclonal activation of B-cells by the Epstein-Barr (EB) virus. They are seen also in lepromatous leprosy where the abundance of mycobacteria reproduces some of the features of Freund's adjuvant. However, unlike the usual situation in human autoimmune disease, these autoantibodies tend to be IgM and, normally, do not persist when the microbial components are cleared from the body. It is likely that the reactions largely involve B-1 cells. Curiously, lymphocytes from many patients with SLE and from mice with spontaneous lupus produce abnormally large amounts of IgM when cultured *in vitro* as if they were under polyclonal activation. Nevertheless, it is difficult to see how a pan-specific polyclonal activation could give rise to the patterns of autoantibodies

AUTOIMMUNITY CAN ARISE THROUGH BYPASS OF REGULATORY MECHANISMS

Regulatory cells try to damp down autoimmunity

It should be emphasized that these T-helper bypass mechanisms for the induction of autoimmunity do not by themselves ensure the continuation of the response, since normal animals have been shown to be capable of damping down autoantibody production through regulatory T-cell interactions as, for example, in the case of red cell autoantibodies induced in mice by injection of rat erythrocytes (figure 19.15). When regulatory T-cell activity is impaired by low doses of cyclophosphamide, or if strains like the SJL which have prematurely aging regulators are used, induced autoimmunity is prolonged and more severe. Yet another example is the protection against autoimmune diabetes and thyroiditis which develops in irradiated adult thymectomized rats by injection of CD4⁺,

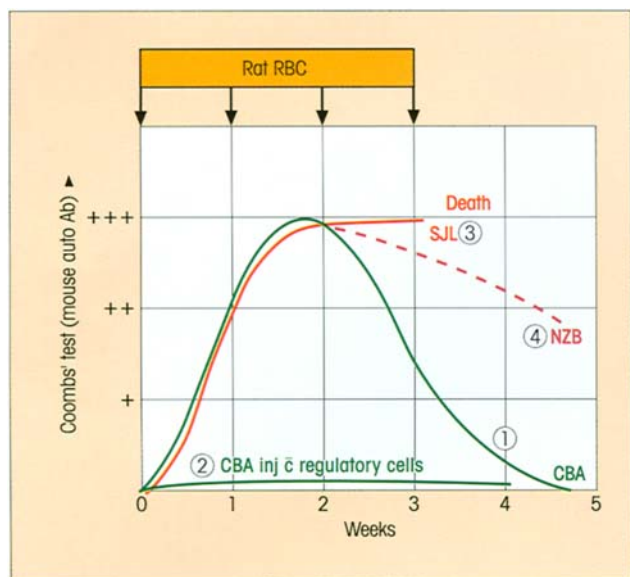


Figure 19.15. Regulation of self-reactivity. When CBA strain mice (1) are injected with rat red cells, autoantibodies are produced by this cross-reacting antigen (see p. 413) which coat the host erythrocytes and are detected by the Coombs' antiglobulin test (see p. 334). Despite repeated injections of rat erythrocytes, the autoantibody response is switched off by the expansion of CD4 mouse red cell-specific regulatory cells which do not affect antibody production to the heterologous erythrocyte determinants. When these regulatory cells are injected into naive CBA mice (2), rat red cells cannot induce autoantibodies. The SJL strain (3), in which suppressor activity declines rapidly with age, is unable to regulate the autoimmune response and develops particularly severe disease. The response is also prolonged in the autoimmune NZB strain (4). (Based on data of Cooke A. & Hutchings P., e.g. *Immunology* 1984, 51, 489.)

CD45RB^{lo}, RT6⁺ T-cells from a normal donor. This phenotype (Tr1) is characteristic of the mucosal regulatory cells which mediate oral tolerance; they produce IL-10 on activation and promote the differentiation of T-cells that secrete TGF β and skew responses towards the Th2-type pole. This notion accords with the finding that the generation of regulatory cells which modulate susceptibility can be influenced by gut flora: Fisher rats raised in a pathogen-free facility are susceptible to adjuvant arthritis (table 19.4) unless moved to conventional surroundings, although this resistance is lost if they are fed the antibiotic, neomycin. Furthermore, resistance can be transferred to susceptible rats by spleen cells stimulated *in vitro* with C-terminal determinants of mycobacterial hsp65. This ties up with suppression of diabetes in NOD mice by injection of a peptide from mammalian hsp. Yet another player in the field is the NK1 T-cell which is deficient in NOD mice, but can prevent the development of diabetes if transferred from F1(Balb/c $\times$ NOD) donors. From quite a different tack, small numbers of an encephalitogenic T-cell clone **vaccinated** normal recipients against the patho-

genic consequences of a subsequent higher dose, hinting strongly at anti-idiotypic control. The interrelationships between this panoply of antigen-, idiotype-, hsp- and possibly nonspecific regulators (figure 19.16) need sorting.

Neonatal thymectomy deletes potential self-regulators

We are surely nodding our grudging assent to the idea that, in general, manipulations which reduce regulatory T-cells encourage the development of autoantibodies. Even so, the effect of thymectomy within a narrow window of 2–4 days after birth in the mouse is quite startling in that it gives rise to widespread organ-specific autoimmune disease affecting mainly the stomach, thyroid, ovary, prostate and sperm; circulating antibodies are frequently detected and deposits of Ig and complement are often seen around the basement membranes. Spleen cells from intact adult males, but not females, injected into these 3-day thymectomized mice can prevent the development of prostatitis, although both are able to prevent gastritis, from which one concludes that the normal male has additional suppressor T-cells specific for prostate and activated by prostate antigens. We have alluded earlier to the evidence for intrathymic expression of mRNA for a whole set of nominally organ-specific antigens such as insulin, thyroglobulin and myelin basic protein. These are expressed in rare large medullary cells often at the center of lymphocyte rosettes, presumably guiding the formation of potential organ-specific suppressive regulators between days 2 and 4, the time at which thymectomy upsets the balance between autoreactive and suppressor cells.

If neonatal thymectomy really does greatly deplete the T-suppressor population, the early induction or exacerbation of spontaneous autoimmune states in susceptible animals—autoimmune hemolytic anemia in NZB mice and thyroiditis in Obese strain chickens and Buffalo rats—is not entirely unexpected.

Defects in regulation contribute to spontaneous autoimmunity

Such would be the conclusion from the demonstration that transfer of disease from splenocytes of recently diabetic NOD mice into NOD-SCID congenics could be inhibited by the CD4, CD45RB^{lo} (memory) splenic subset of young, nondiabetic animals. Along the same lines, it was shown long ago that Coombs' positivity (i.e. the state in which circulating red cells are coated with autoantibody) can be transferred with the spleen cells of a Coombs'-positive NZB mouse to a *young, neg-*

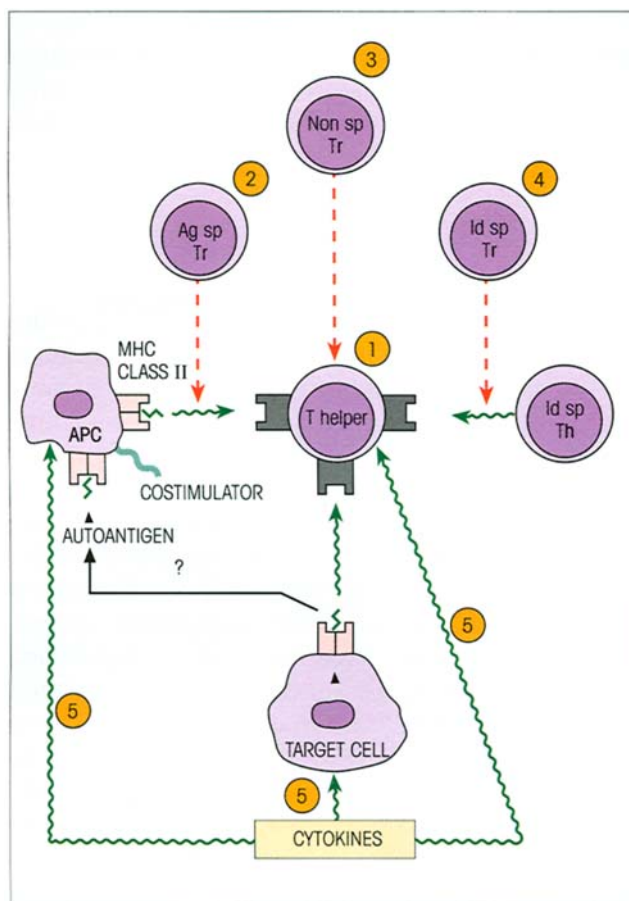


Figure 19.16. Bypass of regulatory mechanisms leads to triggering of autoreactive T-helper cells through defects in (1) tolerizability or ability to respond to or induce T-regulators (Tr), or (2) expression of antigen-specific, (3) hsp and other nonspecific or (4) idiotype-specific T-regulators, or (5) through imbalance of the cytokine network, producing derepression of class II genes with inappropriate cellular expression of class II and presentation of antigen on target cell, stimulation of APC, and possible activation of anergic T-helper. The beneficial effect of pooled whole Ig in certain human autoimmune diseases, such as idiopathic thrombocytopenic purpura, and of T-cell vaccination in experimental autoimmune encephalomyelitis (EAE) and adjuvant arthritis in rats lends weight to the idea of idiotype control mechanisms. Evidence for a regulatory CD4 subset comes from studies reporting inflammatory infiltrates in liver, lung, stomach, thyroid and pancreas in athymic rats reconstituted with CD45RB^{hi}/CD4 T-cells but not with unfractionated or CD45RB^{lo} CD4 cells (presumably the regulators).

active mouse of the same strain, but the continued production of red cell antibodies is short lived unless the recipient's T-cells are first depleted by pretreatment with antilymphocyte serum.

A progressive loss of regulatory cells with age may account for the inability of the NZB mouse to normalize the experimental induction of red cell autoimmunity (figure 19.15) and for the increasing resistance to the induction of tolerance to soluble proteins in elderly NZB mice, apparently associated with a sudden fall in

the plasma concentration of the thymic peptide thymulin before the onset of disease (note: thymulin is said to inhibit the autoreactive response of spleen cells to syngeneic fibroblasts in culture).

Do abnormalities in apoptotic mechanisms contribute to these regulatory defects? T- and B-cells of NOD mice are resistant to apoptosis, as are lymphocytes of the MRL/*lpr* lupus mouse strain which has a *fas* gene mutation. This mutation produces the characteristic lymphoproliferation, and possibly failure to limit the expansion of self-reactive T- and B-cell clones by apoptosis. The *gld* lupus model complements this situation with mutations in the *fas ligand*.

Peripheral T-cell responses are seriously controlled by dendritic cells. The subset of phenotype CD11c⁺/b⁻/8α⁺ favors Th1 cells, while that of CD11c⁺/b⁺/8α⁻ promotes Th2 responses. IL-4 prevents maturation of dendritic cells, and subsequent interaction of these immature cells with cognate T-cells results in peripheral tolerization. CD30/CD30L signaling is important for peripheral deletion of self-reactive T-cells; thus, deletion of pancreatic islet reactive CD8 T-cells failed to occur in CD30 knockout mice and these cells were highly aggressive, as few as 150 provoking diabetes in adoptive recipients. The numbers of dendritic cells in humans at risk for diabetes and in NOD mice are grossly reduced, which would accord with a critical role in preventing disease by promoting autoregulatory responses. Another curious feature is that macrophages from diabetic patients and relatives with the susceptible HLA haplotype constitutively express high levels of the COX-2 enzyme responsible for the synthesis of prostanooids. Something interesting is simmering here and should break surface soon.

We have previously drawn attention to the distinctive properties of the B-1 population with respect to its propensity to synthesize IgM autoantibodies and its possible intimate relationship to the setting up of the regulatory idiotype network (cf. p. 207), and one must seriously entertain the hypothesis that unregulated activity by these cells could be responsible for certain autoimmune disorders. The pitifully named moth-eaten strain is heavily into autoimmunity, and the mice make masses of anti-DNA and anti-polymorphs and die with intense pneumonitis, often before they have tasted the fruits of life. They exhibit reduced catalytic activity of their protein tyrosine phosphatase 1C due to mutation. Their IgM levels rise to a staggering 25–50 times normal and —this is quite bizarre— their B-cells are nearly all CD5⁺, i.e. B-1. This population is also raised in the NZB mouse and largely accounts for the production of the IgM autoantibodies in this strain. Now here is a persuasive experiment. When trans-

genes encoding an NZB red cell autoantibody were introduced into normal mice, no B-2 cells were present and 50% developed autoimmune disease. Intraperitoneal injection of erythrocytes deleted the B-1 cells and prevented disease. This tells us that the NZB hemolytic anemia is due to red blood cell autoantibodies produced by B-1 cells, that this population is only tolerated properly when the antigen gains access to the peritoneum where they develop, and that some (around 50%), but not all, animals can control the autoreactive clones. Whether B-1 cells escape regulatory control and undergo an unrestrained isotype switch to the pathogenic IgG antibodies responsible for disease in other models, such as the NZB $\times$ W mouse, is still a question on which the jury's verdict is awaited, although depletion of B-1 lymphocytes greatly reduces the immune complex glomerulonephritis. The IdD-23 idiotype characteristic of natural autoantibodies has been identified on a monoclonal IgG anti-DNA—but one idiotype doesn't make a summer, if we can misquote a well-known saying!

In humans, a high proportion of B-1 cells make IgM rheumatoid factors (anti-Fc γ) and anti-DNA using germ-line genes. In rheumatoid arthritis patients, although there are increased numbers of circulating B-1 cells, the polyclonal rheumatoid factors synthesized do not, by and large, bear the public idiotypes of this subset. SLE could be different because the 16/6 public idiotype associated with germ-line genes encoding anti-DNA is found on a significant fraction of the IgG anti-DNA in patients' serums. Gene sequencing may be required to establish the relationship between B-1 cells and IgG autoantibody synthesis.

Aside from the defective IL-1/6–hypothalamic feedback loop giving rise to low corticosteroid levels in rheumatoid arthritis described earlier (cf. figure 19.7), less is known of regulatory circuits in humans, although there is increasing evidence that nonspecific T-suppressor function in SLE may be poorly regulated. B-lymphocytes from patients with active disease secrete larger amounts of Ig when cultured *in vitro* than normal B-cells. Concanavalin-A-induced nonspecific suppressors are reduced or absent and T-cells with Fc γ receptors, which suppress pokeweed mitogen-stimulated lymphocytes, are low, the defect being greater the more active the disease. The production of thymulin and of IL-2 is also depressed in these patients. A significant proportion of clinically unaffected close relatives also demonstrate abnormally low levels of nonspecific suppressors, indicating that the deficit in SLE patients is not a consequence of the illness or its treatment and that additional factors must be implicated in the causation of disease.

In any case, it is difficult to account for the antigenic specificity of different autoimmune disorders on the basis of a generalized depression of nonspecific suppressors *alone*, without invoking defects in either antigen- or idiotype-specific suppressor T-cells. There is, however, a further possibility which has aroused much interest.

Upregulation of T-cell interaction molecules

The majority of organ-specific autoantigens normally appear on the surface of the cells of the target organ in the context of class I but not class II MHC molecules. As such they cannot communicate with T-helpers and are therefore immunologically silent. Pujol-Borrell, Bottazzo and colleagues reasoned that, if the class II genes were somehow derepressed and class II molecules were now synthesized, they would endow the surface molecules with potential autoantigenicity (figure 19.16). Indeed, they have been able to show that human thyroid cells in tissue culture can be persuaded to express HLA-DR (class II) molecules on their surface after stimulation with γ -interferon (IFN γ), and, further, that the cytoplasm of epithelial cells from the glands of patients with Graves' disease (thyrotoxicosis) stains strongly with anti-HLA-DR reagents, indicating active synthesis of class II polypeptide chains (figure 19.1b). Inappropriate class II expression has also been reported on the bile ductules in primary biliary cirrhosis and on endothelial cells and some β -cells in the diabetic pancreas both in the human and in the BB rat model.

Whether adventitious expression of class II on these cells through activation by something like virally induced IFN is responsible for *initiating* the autoimmune process by priming autoreactive T-helpers, or whether reaction with *already activated* T-cells induces class II by release of IFN γ and makes the cell a more attractive target for provoking subsequent tissue damage, is still an unresolved issue. However, transfection of mice with the class II *H-2A* genes linked to the insulin promoter led to expression of class II on the β -islet cells of the pancreas but did *not* induce autoimmunity. Lack of B7 costimulatory molecules seems to be responsible for the failure of these class II-positive β -cells to activate naive T-cells, a job which may have to be left to the professional dendritic APCs.

Cytokine imbalance may induce autoimmunity

In contrast, transfection with the IFN γ gene on the insulin promoter under the same circumstances produced a local inflammatory reaction in the pancreas

with aberrant expression of class II *and* diabetes; this must have been a result of autoimmunity since a normal pancreas grafted into the same animal suffered a similar fate. This implies that unregulated cytokine production producing a local inflammatory reaction can initiate autoimmunity, probably by enhancing the presentation of islet antigen by recruiting and activating dendritic cells, by increasing the concentration of processed intracellular autoantigen available to them, and by increasing their avidity for naive T-cells through upregulation of adhesion molecules; perhaps previously anergic cells may be made responsive to antigen (figure 19.16). Once primed, the T-cells can now interact with the islet β -cells which will be displaying increased amounts of class II and adhesion molecules for T-cells on their surface.

This all seems very straightforward but, although other proinflammatory cytokines, IL-12 and TNF as well as IFN γ , can promote the induction of organ-specific autoimmune disease at an early time by priming pathogenic Th1 responses, late expression of the same cytokines can drive the terminal differentiation and death of autoreactive T-cells. Thus we can in fact correct some spontaneous models of autoimmune disease by the injection of cytokines: IL-1 cures the diabetes of NOD mice, tumor necrosis factor (TNF) prevents the onset of SLE symptoms in NZB $\times$ W hybrids and transforming growth factor- β 1 (TGF β 1) is known to protect against collagen arthritis and relapsing experimental autoimmune encephalomyelitis (EAE). The pleiotropic effects of the cytokines on different cell types involved at different stages in these diseases, and their positive and negative networking interactions with each other, add some uncer-

tainty to the analysis and prediction of these complex events.

AUTOIMMUNE DISORDERS ARE MULTIFACTORIAL

We must come back to this. Undoubtedly, the autoimmune diseases have a multifactorial etiology combining polygenic traits and environmental influences. Many of the defects we have discussed, individually not necessarily uncommon, may contribute in various combinations to different disorders. No one gene is sufficient or required for disease onset and in any individual, disease susceptibility, presentation in terms of target organ severity and prognosis reflect additive or epistatic effects of several fortuitously inherited alleles, many of which may be shared with other autoimmune diseases. Even in a disease-prone strain of mice expressing an identical array of susceptibility genes, the proportion of animals developing autoimmune pathology (penetrance) increases with age and suggests that the expression of these complex traits requires an internal stochastic or environmental triggering of events, the probability of which increases with time. Thus, superimposed upon a genetically complex susceptibility, we might be dealing with some aging process affecting the thymus or the lymphoid stem cells and their internal control of self-reactivity. Sex hormones and defective pituitary–adrenal feedback loops may contribute. Now throw into this melange a panoply of dietary and other environmental factors, particularly microbial agents, which could have a variety of effects on the target organs, the lymphoid system and the cytokine network.

SUMMARY

The immune system balances precariously between effective responses to environmental antigens and regulatory control of an array of potentially suicidal responses to self-molecules.

The scope of autoimmune diseases

- Autoimmunity is associated with certain diseases which form a spectrum. At one pole, exemplified by Hashimoto's thyroiditis, the autoantibodies and the lesions are **organ-specific** with the organ acting as the target for autoimmune attack; at the other pole are the **nonorgan-specific** or **systemic autoimmune diseases**, such as SLE,

where the autoantibodies have widespread reactivity and the lesions resemble those of serum sickness relating to deposition of circulating immune complexes.

- There is a tendency for organ-specific disorders such as thyroiditis and pernicious anemia to overlap in given individuals, while overlap of rheumatological disorders is greater than expected by chance.
- There are a number of models of organ-specific and systemic autoimmune diseases which occur spontaneously (e.g. nonobese diabetic mice or NZB $\times$ W hybrids with SLE) or can be induced experimentally (e.g. thyroiditis by thyroglobulin in complete Freund's adjuvant (CFA) and

(continued p. 420)

perhaps SLE by immunization with Id⁺ anti-DNA monoclonal in CFA).

Genetic and environmental influences

- Multifactorial genetic factors increase predisposition to autoimmune disease: these include HLA tissue type, the predisposition to aggressive autoimmunity and the selection of potential autoantigens.
- Females have a far higher incidence of autoimmunity than males, perhaps due to hormonal influences.
- Feedback control of lymphocytes through the cytokine–hypothalamus–pituitary–adrenal loop may be defective as shown for rheumatoid arthritis.
- Twin studies indicate a strong environmental influence in many disorders; both microbial and nonmicrobial factors have been suspected.

Autoreactivity comes naturally

- B-1 cells form a pool of mutually stimulating cells spontaneously producing ‘natural antibodies’ which interact idiotypically and frequently show multispecific autoreactivity.
- The immune system appears to have a set of T-cells directed to a limited number of dominant autoantigens which are tightly controlled.

Is autoimmunity driven by antigen?

- In spontaneous models of diabetes and thyroiditis, removal of antigen prevents autoimmunity.
- The development of high affinity mutated antibodies and immune responses to clusters of anatomically related antigens strongly imply B-cell selection of autoantigen.
- T-cell specificities in systemic autoimmunity are unknown but may be anti-idiotypic.
- Autoantigens are, for the most part, accessible to circulating lymphocytes which normally include autoreactive T- and B-cells. Dominant autoantigens will induce tolerance but T-cells specific for peptides presented at low concentrations (cryptic epitopes) will be potentially autoreactive.

The T-helper is pivotal for control

- It is assumed that the key to the system is the control of autoreactive T-helper cells which are normally unresponsive because of clonal deletion, clonal anergy, T-suppression or inadequate autoantigen processing.

Autoimmunity can arise through bypass of T-helpers

- Abnormal modification of the autoantigen, cross-reaction with exogenous antigens or ‘piggy-back’ recognition of T-helper epitopes can provide new carrier determinants and epitope spread.
- T-helpers could also be bypassed by idiotypic network interactions with cross-reactions between public idiotypes on autoantibodies and microbial antibodies or microbes themselves, or by antibodies formed to antiviral idiotypes which behaved as internal images of the virus and reacted with the cell surface viral receptor.
- Finally, B-cells and T-cells can be stimulated directly by polyclonal activators such as EB virus or superantigens.

Autoimmunity can arise through bypass of regulatory mechanisms

- T-helper bypass alone may be insufficient to *maintain* autoimmunity and it is generally considered that, in addition, a defect in cells which normally regulate autoimmunity is required.
- This could occur through an inability of the central T-helper cell to be tolerized or to respond to or induce T-suppressors.
- It could also arise through defects in antigen-specific, idiotypic-specific and hsp and perhaps other nonspecific T-suppressor systems.
- Another possibility would be the derepression of class II genes giving rise to inappropriate cellular expression of class II so breaking the ‘silence’ between cellular autoantigen and autoreactive T-inducer.
- This would make the cell a target for activated T-cells but, without costimulators such as B7, perhaps only professional APCs could prime the resting autoreactive T-cell.
- Cytokine imbalance provides the circumstances for this to occur although the situation may be very complex.

Autoimmune disorders are multifactorial

- Given the polygenic predisposition, these changes could come about by some spontaneous internal dysregulation related perhaps to aging, and/or through environmental factors, particularly microbes, which could act in an uncomfortably large number of different ways.

See the accompanying website (www.roit.com) for multiple choice questions.

FURTHER READING

Suggestions for further reading are given at the end of Chapter 20.

Autoimmune diseases

2 — Pathogenesis, diagnosis and treatment

Introduction, 421

Pathogenic effects of humoral autoantibody, 421

Blood cells, 421

Surface receptors, 422

Other tissues, 423

Pathogenic effects of complexes with autoantigens, 425

Systemic lupus erythematosus (SLE), 425

Rheumatoid arthritis, 427

T-cell-mediated hypersensitivity as a pathogenic factor in autoimmune disease, 431

Rheumatoid arthritis again, 431

Organ-specific endocrine disease, 433

Multiple sclerosis (MS), 435

Psoriasis, 436

Some other systemic vascular disorders with immunopathological components, 436

Diagnostic value of autoantibody tests, 438

Treatment of autoimmune disorders, 438

Control at the target organ level, 438

Anti-inflammatory drugs, 440

Immunosuppressive drugs, 441

Immunological control strategies, 442

Summary, 446

Further reading, 449

INTRODUCTION

We should now look at the evidence which helps to uncover the mechanisms by which autoimmunity, however it arises, plays a **primary pathogenic role** in the production of tissue lesions within the group of diseases labeled as 'autoimmune'.

PATHOGENIC EFFECTS OF HUMORAL AUTOANTIBODY

Blood cells

The erythrocyte antibodies play a dominant role in the destruction of red cells in **autoimmune hemolytic anemia**. Normal red cells coated with autoantibody eluted from Coombs' positive erythrocytes (cf. figure 16.16) have a shortened half-life after reinjection into the normal subject, essentially as a result of their adherence to Fcγ receptors on phagocytic cells in the spleen. Remember also that B-1 cells in a mouse bearing a transgene encoding the New Zealand Black (NZB) mouse red cell autoantibody cause hemolytic disease (cf. p. 418). To put the case for B-1 cells beyond all reasonable doubt, any manipulation of NZB mice which elimi-

nates these cells, such as intraperitoneal injection of water, treatment with anti-interleukin (IL)-10 or introduction of the *xid* gene, prevents the development of hemolytic anemia.

Some children with immunodeficiency associated with very low white cell counts have a serum lymphocytotoxic factor which requires complement for its activity. Lymphopenia occurring in patients with systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) may also be a direct result of antibody, since nonagglutinating antibodies coating the white cells have been reported in such cases.

Platelet antibodies are apparently responsible for **idiopathic thrombocytopenic purpura** (ITP). IgG from a patient's serum when given to a normal individual causes a depression of platelet counts and the active principle can be absorbed out with platelets. The transient neonatal thrombocytopenia which may be seen in infants of mothers with ITP is explicable in terms of transplacental passage of IgG antibodies to the child.

The primary **antiphospholipid syndrome** is characterized by recurrent venous and arterial thromboembolic phenomena, recurrent fetal loss, thrombocytopenia and cardiolipin antibodies. Passive transfer of such antibodies into mice is fairly devastating,

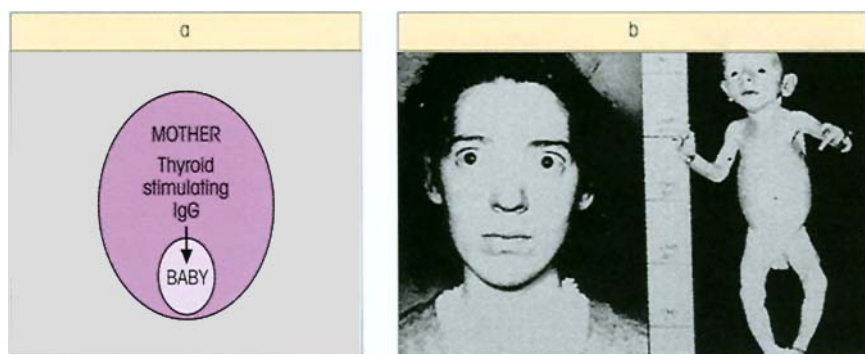


Figure 20.1. Neonatal thyrotoxicosis. (a) The autoantibodies which stimulate the thyroid through the TSH receptors are IgG and cross the placenta. (b) The thyrotoxic mother therefore gives birth to a baby with thyroid hyperactivity which spontaneously resolves as the mother's IgG is catabolized. (Photograph courtesy of Professor A. MacGregor.)

resulting in lower fecundity rates and recurrent fetal loss. The effect seems to be mediated through reaction of the autoantibodies with a complex of cardiolipin and β_2 -glycoprotein 1 which inhibits triggering of the coagulation cascade. The placental trophoblast is a primary target of these antibodies since the villous cytotrophoblast is one of the few cell types which externalizes phosphatidyl serine during development.

Surface receptors

Thyroid

Under certain circumstances antibodies to the surface of a cell may stimulate rather than destroy (cf. 'stimulating hypersensitivity'; Chapter 16). This would seem to be the case in **Graves' disease** (thyrotoxicosis or Basedow's disease) where a direct link with autoimmunity came with the discovery by Adams and Purves of thyroid-stimulating activity in the serum of these patients (Milestone 19.1), ultimately shown to be due to the presence of antibodies to TSH receptors (TSH-Rs), which seem to act in the same manner as TSH itself (cf. p. 345). Both operate through the adenyl cyclase system as indicated by the potentiating effect of theophylline, and both produce similar changes in ultrastructural morphology in the thyroid cell, but it is one of Nature's 'passive transfer experiments' which links TSH-R antibodies most directly with the pathogenesis of Graves' disease. When thyroid-stimulating antibodies (TSABs) from a thyrotoxic mother cross the placenta, they cause the production of neonatal hyperthyroidism (figure 20.1), which resolves after a few weeks as the maternal IgG is catabolized.

There is a good correlation between the titer of TSAB and the severity of hyperthyroidism. Because TSABs act independently of the pituitary–thyroid axis, iodine

uptake by the gland is unaffected by the administration of thyroxine or tri-iodothyronine, whereas normally this would cause feedback inhibition and suppression of uptake; this forms the basis of an important diagnostic test for thyrotoxicosis.

There is reason to believe that enlargement of the thyroid in this disorder is due to the action of antibodies which react with a 'growth' receptor and directly stimulate cell division as distinct from metabolic hyperactivity. In contrast, sera from patients with **primary myxedema** (atrophic thyroiditis) contain antibodies capable of blocking the stimulation of growth by TSH, thereby preventing the regeneration of follicles which is a feature of the enlarged **Hashimoto goiter**. Graves' disease is often associated with exophthalmos which might be due to cross-reaction of antibodies to a 64 kDa membrane protein present on both thyroid and eye muscle.

Muscle and nerve

The transient muscle weakness seen in a proportion of babies born to mothers with **myasthenia gravis** calls to mind neonatal thrombocytopenia and hyperthyroidism and would certainly be compatible with the transplacental passage of an IgG capable of inhibiting neuromuscular transmission. Strong support for this view is afforded by the consistent finding of antibodies to muscle acetylcholine receptors (ACh-Rs) in myasthenics and the depletion of these receptors within the motor endplates. In addition, myasthenic symptoms can be induced in animals by injection of monoclonal antibodies to ACh-R or by active immunization with the purified receptors themselves. Nonetheless, the majority of babies with myasthenic mothers do not display muscle disease and it may be that they are protecting themselves through production of antibodies directed to idiotypes on the maternal autoantibodies.

Many myasthenics develop thymomas bearing molecules which cross-react with ACh-R although structurally not belonging to that gene family. It must be on the cards that molecular mimicry based on comparable peptide sequences may prime autoreactive T-cells which then drive genuine anti-ACh-R responses. An association with ACh-R polymorphism hints at some contribution to risk from the autoantigen.

Neuromuscular defects can also be elicited in mice injected with serum from patients with the **Lambert-Eaton** syndrome containing antibodies to presynaptic calcium channels. Autoantibodies to sodium channels which cross-react with *Campylobacter* bacilli have been identified in **Guillain-Barré syndrome**, a self-resolving peripheral polyneuritis. Rather more way out is **Rasmussen's encephalitis**, a childhood disease of relentless and intractable focal seizures with an inflammatory histopathology in the brain; these patients have antibodies which act as agonists and kill kainic acid-responsive neurons through overstimulation of type 3 glutamine receptors. Would one uncover yet more phenomena of this kind in other neurological disorders if the search was widened and intensified?

Stomach

The underlying histopathological lesion in **pernicious anemia** is an atrophic gastritis in which a chronic inflammatory mononuclear invasion is associated with degeneration of secretory glands and failure to produce gastric acid. The development of achlorhydria is almost certainly accelerated by the inhibitory action of antibodies to the gastric proton pump, an H^+ , K^+ -dependent ATPase located in the membranes of the secretory canaliculi, and possibly also the gastrin receptors.

The idea that some cases of gastric ulcer may result from the stimulation of acid secretion by activation through antibodies to histamine receptors is appealing and we still await the further work required to establish its validity.

Other cellular receptors

Some patients with **atopic allergy** have serum blocking antibodies to β -adrenergic receptors and these may represent just one of many different types of factor which could alter the baseline sensitivity of mast cells and make the individual more at risk for the development of disease. The flip side of the coin is revealed in the cardiomyopathy of Chagas' disease where antibodies to these receptors act as agonists and increase the heart rate. Antibodies which block insulin recep-

tors are a rare exotic species found in patients with acanthosis nigricans (type B) and ataxia telangiectasia associated with insulin resistance.

Other tissues

Gut

Some patients with **autoimmune atrophic gastritis** diagnosed by achlorhydria and parietal cell antibodies (see above and table 19.2) just meander on year after year without developing the vitamin B_{12} deficiency which precipitates **pernicious anemia**. It is probable that autoallergic destruction is roughly balanced by regeneration of mucosal cells, an explanation which could account for the observation that high doses of steroids may restore gastric function in certain patients with pernicious anemia. However, the balance would be upset were the patient now to produce antibodies to intrinsic factor in the lumen of the gastrointestinal tract; these would neutralize the small amount of intrinsic factor still available and the body would move into negative balance for B_{12} . The symptoms of B_{12} deficiency, pernicious anemia and sometimes subacute degeneration of the cord, would then appear some considerable time later as the liver stores became exhausted (figure 20.2).

The normally acquired tolerance to dietary proteins seems to break down in **celiac disease** where T-cell sensitivity to wheat gluten in the small intestine can be demonstrated. Since gluten can bind strongly to the extracellular matrix protein, endomysium, one could hypothesize that uptake of the complex by IgA B-cells specific for endomysium would 'piggy-back' the gluten into the B-cell for processing and presentation on MHC class II to gluten-specific T-helpers (cf. figure 19.13). Stimulation of the B-cell would now follow with secretion of the IgA endomysial antibodies which are exclusive to patients with celiac disease. Together with the increased expression of $Fc\alpha$ receptors in the lamina propria and evidence of complement and eosinophil activation, it is conceivable that antibody-mediated mechanisms could be pathogenic.

Skin

An antibody pathogenesis for **pemphigus vulgaris** is favored by the recognition of a 130 kDa autoantigen on stratified squamous epithelial cells which is a member of the cadherin family of Ca^{2+} -dependent adhesion molecules. Likewise, antibodies to desmoglein 1 probably mediate the blistering of the epidermis in **pemphigus foliaceus**.

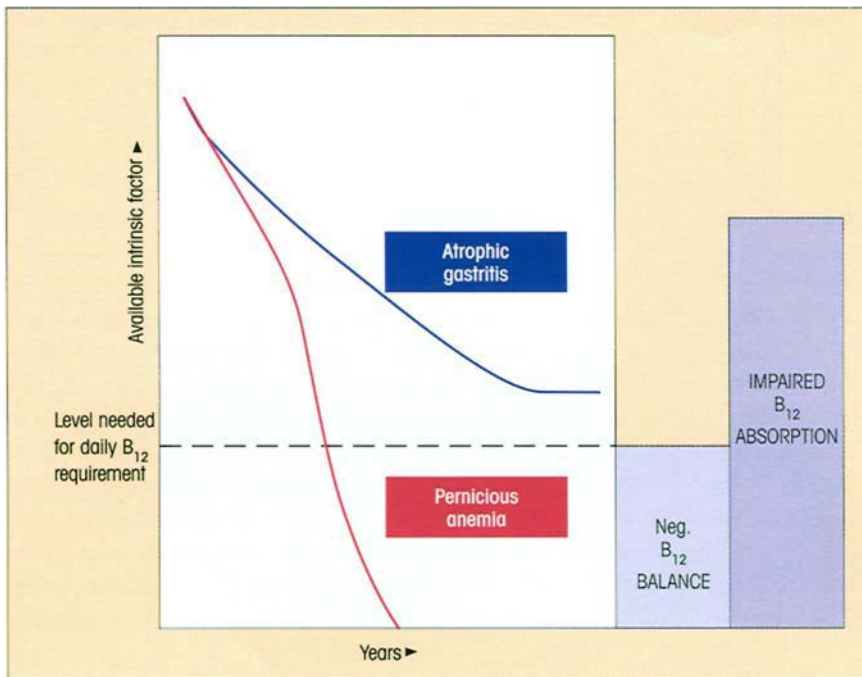


Figure 20.2. Pathogenesis of pernicious anemia. Patients with long-standing atrophic gastritis having parietal cell but no intrinsic factor antibodies do not go into negative B_{12} balance. Pernicious anemia develops when intrinsic factor antibodies become superimposed upon the atrophic gastritis. (After Doniach D. & Roitt I.M. (1964) *Seminars in Hematology* I, 313.)

Sperm

In some **infertile males**, agglutinating antibodies cause aggregation of the spermatozoa and interfere with their penetration into the cervical mucus.

Glomerular basement membrane (g.b.m.)

With immunological kidney disease, the experimental models preceded the finding of parallel lesions in the human. Injection of cross-reacting heterologous g.b.m. preparations in complete Freund's adjuvant produces glomerulonephritis in sheep and other experimental animals. Antibodies to g.b.m. can be picked up by immunofluorescent staining of biopsies from nephritic animals with anti-IgG. The antibodies are largely, if not completely, absorbed out by the kidney *in vivo* but they appear in the serum on nephrectomy and can passively transfer the disease to another animal of the same species.

An entirely analogous situation occurs in humans in certain cases of glomerulonephritis, particularly those associated with lung hemorrhage (**Goodpasture's syndrome**). Kidney biopsy from the patient shows *linear* deposition of IgG and C3 along the basement membrane of the glomerular capillaries (figure 16.17a). After nephrectomy, g.b.m. antibodies can be detected in the serum. Lerner and his colleagues eluted

the g.b.m. antibody from a diseased kidney and injected it into a squirrel monkey. The antibody rapidly fixed to the g.b.m. of the recipient animal and produced a fatal nephritis (figure 20.3). It is hard to escape the conclusion that the lesion in the human was the direct result of attack on the g.b.m. by these complement-fixing antibodies. The lung changes in Goodpasture's syndrome are attributable to cross-reaction with some of the g.b.m. antibodies.

Curiously, mercuric chloride produces anti-g.b.m. glomerulonephritis in Brown Norway rats and, *pari passu*, as the disease remits, there is an upsurge in anti-idiotypic suppressors. Nonsusceptible strains produce suppressors rather promptly.

Heart

Neonatal lupus erythematosus is the most common cause of permanent **congenital complete heart block**. Almost all cases have been associated with high maternal titers of anti-La/SS-B or anti-Ro/SS-A. The mother's heart is unaffected. The key observation was that anti-Ro bound to neonatal rather than adult cardiac tissue and altered the transmembrane action potential by inhibiting repolarization (figure 20.4). IgG anti-Ro reaches the fetal circulation by transplacental passage but, although maternal and fetal hearts are exposed to the autoantibody, only the latter is affected. Anti-

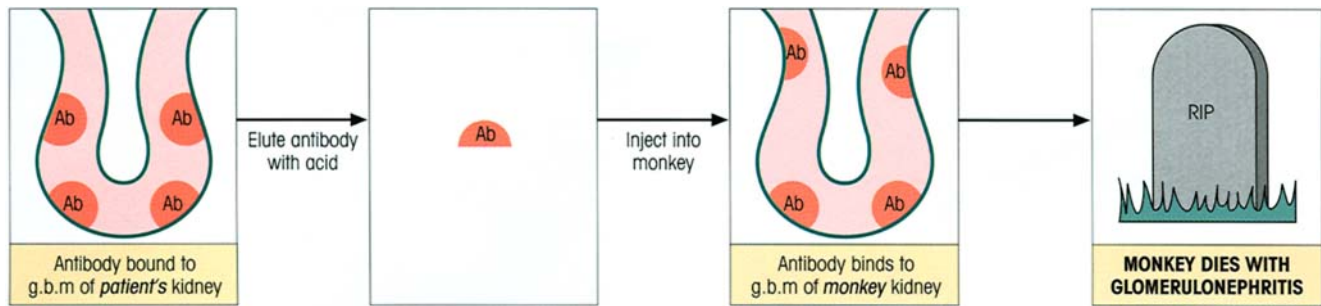


Figure 20.3. Passive transfer of glomerulonephritis to a squirrel monkey by injection of antiglomerular basement membrane (anti-g.b.m.) antibodies isolated by acid elution from the kidney of a patient with Goodpasture's syndrome. (After Lerner R.A., Glascock R.J. & Dixon F.J. (1967) *Journal of Experimental Medicine* 126, 989.)

La/SS-A also binds to affected fetal hearts reacting with laminin in the basement membrane.

PATHOGENIC EFFECTS OF COMPLEXES WITH AUTOANTIGENS

Systemic lupus erythematosus (SLE)

Where autoantibodies are formed against soluble components to which they have continual access, complexes may be formed which can give rise to lesions similar to those occurring in serum sickness, especially when defects in the early classical complement components prevent effective clearance. Thus, although homozygous complement deficiency is a rare cause of SLE (cf. figure 16.24), the archetypal immune complex disorder, it represents the most powerful disease susceptibility genotype so far identified; more than 80% of cases with homozygous C1q and C4 deficiency have SLE. Up to one-half of the patients carry autoantibodies to the collagenous portion of C1q, but in truth there are a rich variety of different autoantigens in lupus (cf. table 19.2), some of them constituents of the nucleosome (cf. figure 19.1g), with the most pathomnemonic being **double-stranded DNA (dsDNA)**. Anti-dsDNA is enriched in cryoglobulins and acid eluates of renal tissue from patients with lupus nephritis where it can be identified, presumably in complexes containing complement, by immunofluorescent staining of kidney biopsies from patients with evidence of renal dysfunction. The staining pattern with a fluorescent anti-IgG or anti-C3 is punctate or 'lumpy-bumpy' as once described (figure 16.17b), in marked contrast with the linear pattern caused by the g.b.m. antibodies

in Goodpasture's syndrome (figure 16.17a; p. 335). The complexes grow in size to become large aggregates visible in the electron microscope as amorphous humps on both sides of the g.b.m. (figure 20.5). During the active phase of the disease, serum complement levels fall as components are affected by immune aggregates in the kidney and circulation. Deposition of complexes is widespread as the name implies and, although 40% of patients *eventually* develop kidney lesions, the corresponding figures for organ involvement are 98% for skin (figure 20.6), 98% for joints/muscle, 64% for lung, 60% for blood, 60% for brain and 20% for heart.

Spontaneous production of anti-dsDNA is also a dominant feature of the animal models of SLE, NZB×W, MRL/*lpr*, BXSB and the *p21* single gene knockout mice, which involve fatal immune complex disease. Cationic anti-DNA, with arginines strategically positioned in locations of paratopic significance, emerges strongly as the disease progresses. The high affinity and IgG class of these antibodies, and the amelioration of symptoms and reduction of renal glomerular immune complexes by treatment of NZB×W mice with DNase I or anti-CD4, provide convincing evidence for a T-dependent antigen-driven complex-mediated pathology. But since DNA itself is not a thymus-dependent antigen and the SLE autoantibodies include a cluster directed to the physically linked antigens constituting the nucleosome, one might envisage a 'piggy-back' mechanism of the type portrayed in figure 19.13. Knowing that nucleosome 'blebs' appear on the surface of apoptotic cells and that a spontaneous expansion of nucleosome-specific T-cell populations precedes the clinical onset of SLE, a likely scenario would involve the internalization of nucleosome material captured on the surface receptors of anti-DNA B-cells, presentation of processed histone peptide–MHC class II complex to the histone-specific T-helper cells, and clonal proliferation of DNA antibody-forming cells (figure 20.7). Complexes of anti-DNA with circulating nucleosome material are demonstrable, and

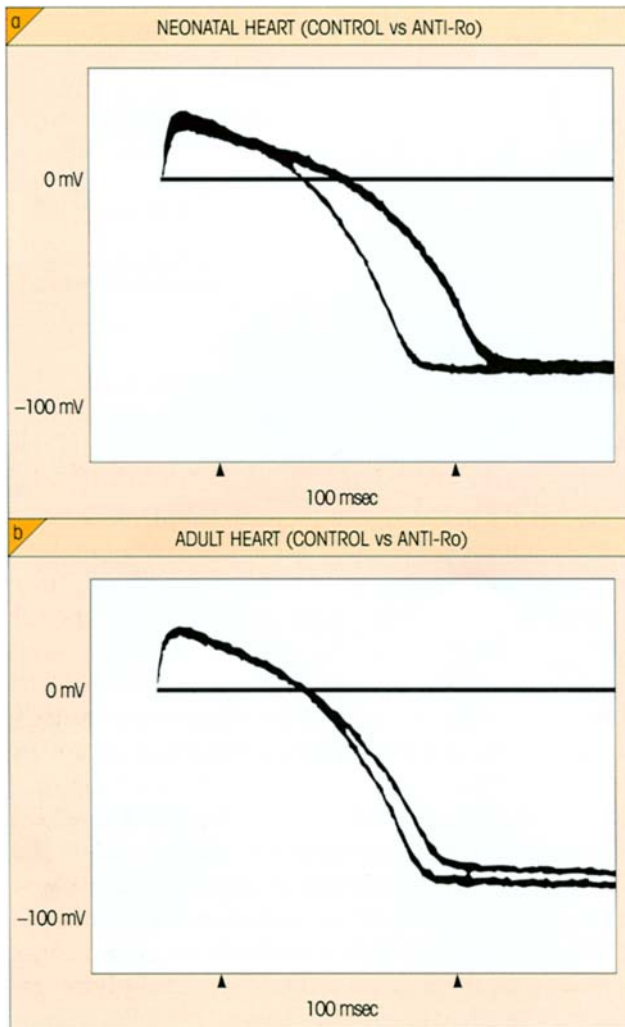


Figure 20.4. Anti-Ro affects conduction in neonatal but not adult heart. (a) Action potential of neonatal NZW rabbit cardiac fiber before and 20 minutes after superfusion with serum containing anti-Ro/SS-A; the repolarization phase of the action potential is reduced by 30%. (b) The same with an adult cardiac fiber showing only 5% reduction. (Reproduced from Alexander E. *et al.* (1992) *Arthritis and Rheumatism* 35, 176.) Anti-La/SS-B can be eluted from the fetal cardiac tissue of infants with congenital heart block. It reacts with fetal but not adult laminin in the basement membrane. Autoantibodies to the endogenous retroviral envelope protein (ERV-3) appear during pregnancy and disappear after delivery, except in women with autoimmune diseases where they persist, the highest levels being found in women with a history of congenital heart block children.

these will bind through the histone (and presumably cationic anti-DNA) to extracellular heparan sulfate where they can accumulate and damage end-organ targets such as the kidney glomerulus.

There is yet another 'piggy-back' pathway. The possible involvement of idiotypes was mooted in the last chapter by reference to experiments in which immunization of mice with human monoclonal antinuclear

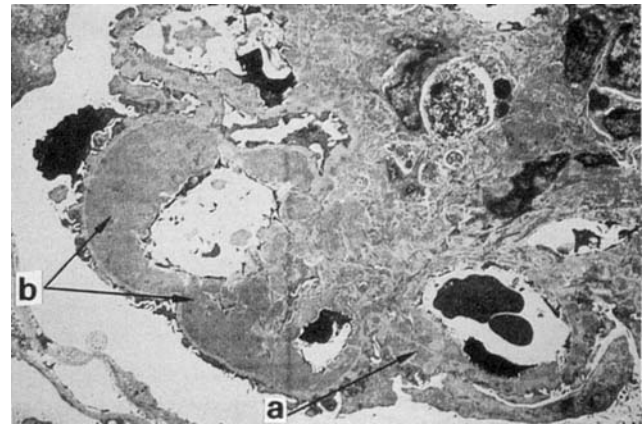


Figure 20.5. Renal biopsy of an SLE patient with severe immune complex glomerulonephritis and proteinuria. Electron micrograph showing irregular thickening of glomerular capillary walls by subepithelial complexes (a) and subendothelial complexes (b). The mesangial region shows abundant (probably phagocytosed) complexes. (Courtesy of Dr A. Leatham.)

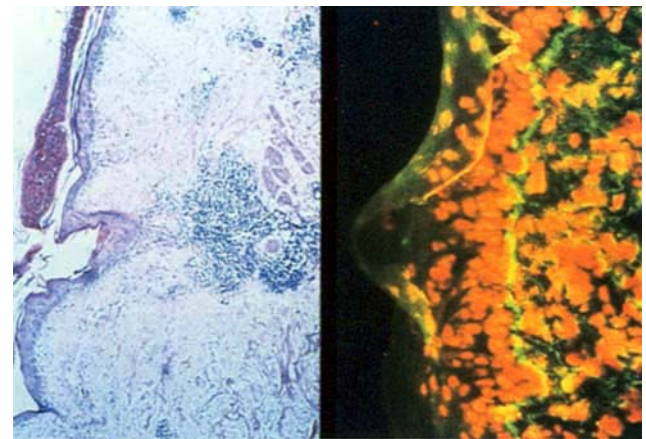


Figure 20.6. The 'lupus band' in SLE. Left—section of skin showing slight thickening of the dermo-epidermal junction with underlying scattered inflammatory cells and a major inflammatory focus in the deeper layers. Low power H & E. Right—green fluorescent staining of a skin biopsy at higher power showing deposition of complexes containing IgG (anti-C3 gives the same picture) on the basement membrane at the dermo-epidermal junction. (Kindly provided by Professor D. Isenberg.)

antibodies gave rise to the production of new antibodies of similar idio type and specificity—in biblical terms, 'antibody begets antibody'. Suppose a major public idio type network is kicked into action by a microbial infection (cf. figure 19.14); for example, the 16/6 idio type on a human anti-DNA circulating as a natural autoantibody is also carried on a germ-line antibody to *Klebsiella*. A T-helper cell recognizing

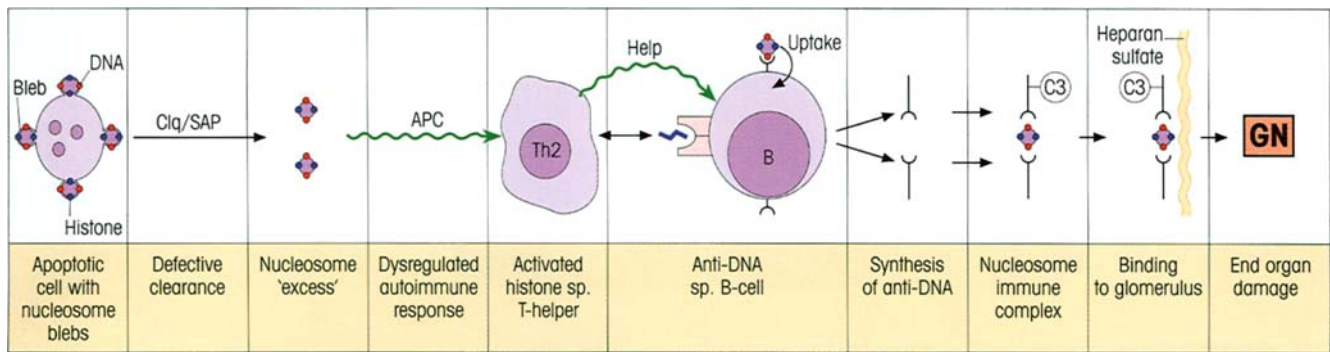


Figure 20.7. Conceivable pathogenetic pathway leading to end-organ damage in SLE. Nucleosomes derived from apoptotic cells can stimulate anti-DNA production by a 'piggy-back' mechanism in susceptible hosts. The resulting complexes bind to heparan sulfate in the glomerular basement membrane where they induce glo-

merulonephritis. The high incidence of lupus in Clq deficient individuals and the susceptibility of lupus patients to skin rashes on exposure to UV in sunlight, which induces apoptosis in skin cells, are well known. SAP, serum amyloid precursor; APC, antigen-presenting cell; GN, glomerulonephritis.

processed 16/6 Id could stimulate anti-DNA B-cells which had captured a complex of DNA with the Id-positive natural autoantibody (cf. figure 19.13 and p. 208).

Rheumatoid arthritis

Morphological evidence for immunological activity

The joint changes in RA are in essence produced by the **malign growth of the synovial cells** as a pannus overlaying and destroying cartilage and bone (figure 20.8a–f). The synovial membrane which surrounds and maintains the joint space becomes intensely cellular as a result of considerable immunological hyper-reactivity, as evidenced by large numbers of T-cells, mostly CD4, in various stages of activation, usually associated with dendritic cells and macrophages (figure 20.8l); clumps of plasma cells are frequently observed and sometimes even secondary follicles with germinal centers are present as though the synovium had become an active lymph node (figure 20.8g–i). Indeed, it has been estimated that the synthesis of immunoglobulins by the synovial tissue ranks with that of a stimulated lymph node. There is widespread expression of surface HLA-DR (class II); T- and B-cells, dendritic and synovial lining cells and macrophages are all positive, indicative of some pretty lively action (figure 20.8k). The thesis is that this fiery immunological reactivity provides an intense stimulus to the synovial lining cells which undergo a Dr Jekyll to Mr Hyde transformation into the invasive pannus which brings about joint erosion through the release of destructive mediators.

IgG autosensitization and immune complex formation

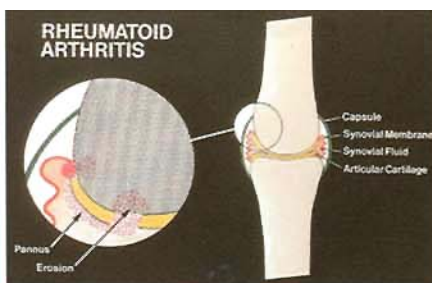
Autoantibodies to the IgG Fc region (see figure 20.10a), known as **antiglobulins** or **rheumatoid factors**, are the hallmark of the disease, being demonstrable in virtually all patients with RA. The majority have IgM antiglobulins which react in the classical latex and sheep cell agglutination tests (table 19.2; note 7), and both they and the 'seronegative' patients who fail to react in these tests can be shown to have elevated levels of **IgG antiglobulins** detectable by solid-phase immunoassay (cf. p. 429; figure 20.9).

If, therefore, autosensitization to IgG is an almost universal feature of the disease, one might expect plasma cells in the synovium to be synthesizing antiglobulins. In fact 10–20% bind fluoresceinated IgG, either in the form of heat-aggregated material (figure 20.8j) or immune complexes (rheumatoid factor is a low affinity antibody and good binding is only seen when multivalent IgG is used as antigen). We must take into account a strange and unique feature of IgG antiglobulins; because they are both antigen and antibody at the same time, they are capable of **self-association** (figure 20.10b) and this hides the majority of free antiglobulin valencies. Cleverly realizing that destruction of the Fc regions by pepsin would liberate these hidden binding sites, Munthe and Natvig observed that a greater percentage of the plasma cells in the synovium displayed an anti-IgG specificity following treatment with this enzyme.

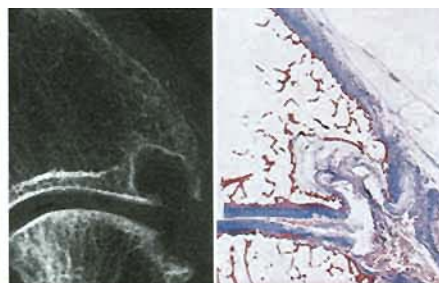
IgG aggregates, presumably products of these plasma cells, can be regularly detected in the synovial tissues and in the joint fluid where they give rise to typical acute inflammatory reactions with fluid exudates.



(a)



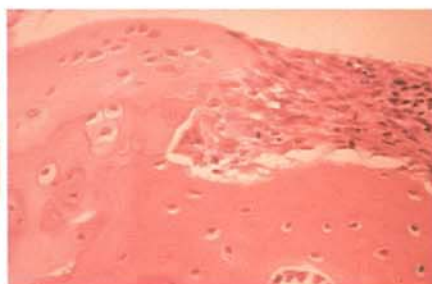
(b)



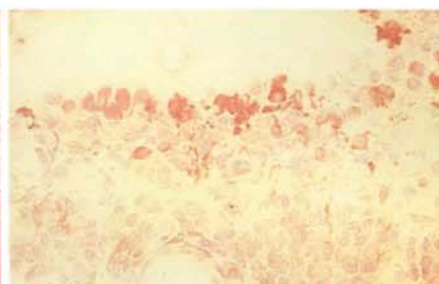
(c)



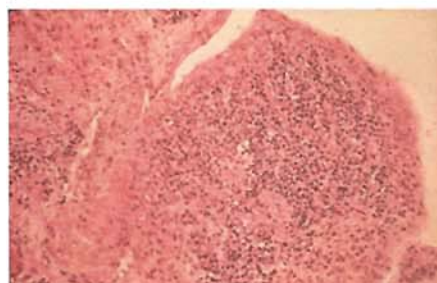
(d)



(e)



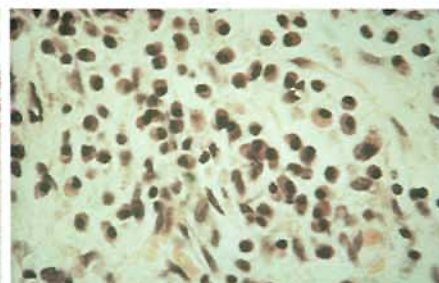
(f)



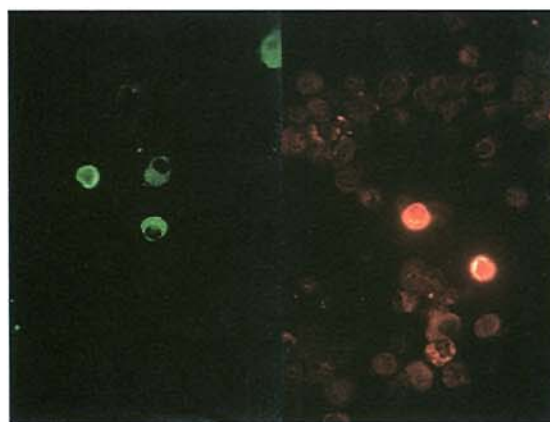
(g)



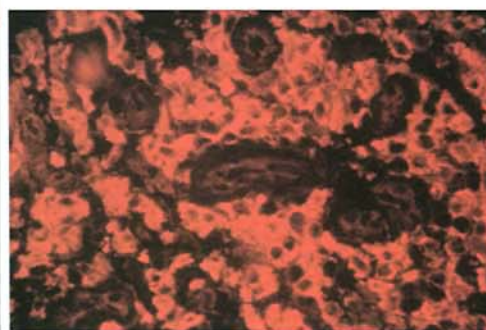
(h)



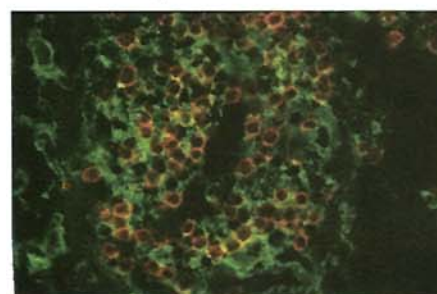
(i)



(j)



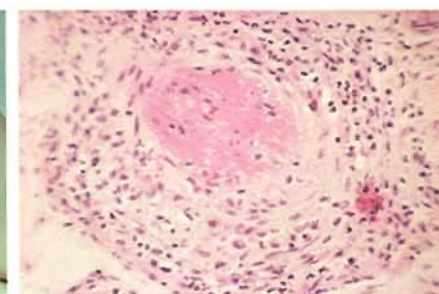
(k)



(l)



(m)



(n)

Analysis shows them to consist almost exclusively of immunoglobulins and complement, while a major proportion of the IgG is present as self-associated antiglobulin as shown by binding to an Fc γ immunosorbent after treatment with pepsin.

Abnormal patterns of IgG glycosylation

The two C_{H2} domains in the Fc region are held apart (cf. p. 48) by two asparagine-linked sugars of the gen-

eral structure shown in figure 20.11a and the terminal galactose lies in a special 'lectin-like' pocket (figure 20.11b). Some chains end in N-acetylglucosamine and lack the terminal galactose sugars. What is extraordinary is that the percentage of sugars completely lacking galactose in the IgG of both juvenile and adult RA patients is nearly always higher than in the controls and can be as high as 60%. This abnormal glycosylation could have four possible consequences.

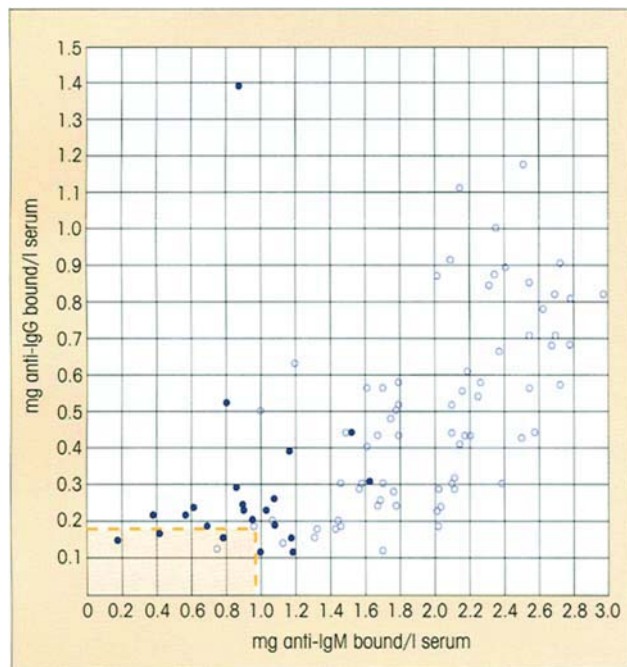


Figure 20.9. IgM and IgG antiglobulins determined by tube radioassay in patients with seropositive (open circles) and seronegative (filled circles) rheumatoid arthritis. The dashed lines indicate the 95% confidence limits (mean $\pm$ 2 SD) of the normal group. (From Nineham L., Hay F.C. & Roitt I.M. (1976) *Journal of Clinical Pathology* 29, 1121.)

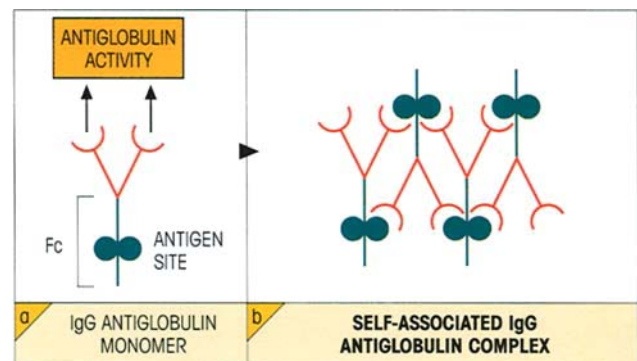


Figure 20.10. Self-associated complexes of IgG antiglobulins. Although of relatively low affinity, the strength of binding is boosted by the 'bonus effect' of the mutual attachment and, furthermore, such complexes in the joint may be stabilized by IgM antiglobulin and C1q which have polyvalent binding sites for IgG. Degradation of the Fc regions by pepsin releases the 'hidden' binding sites involved in the self-association. X-ray analysis of a complex of IgG Fc with two Fab fragments, isolated from a monoclonal IgM rheumatoid factor derived from an RA patient, showed binding of the Fab paratopes using the outer rather than the inner residues of the conventional antigen combining site CDRs; this suggests a novel form of cross-reactivity by allowing simultaneous binding to another antigen. (Sutton B. *et al.* (2000) *Immunology Today* 21, 177.)

Figure 20.8. (Opposite) Rheumatoid arthritis (RA). (a) Hands of a patient with chronic RA showing classical swan-neck deformities. (b) Diagrammatic representation of a diarthrodial joint showing bone and cartilaginous erosions beneath the synovial membrane-derived pannus. (c) Proximal interphalangeal joint depicting marked bony erosion and marginal erosion of the cartilage. (d) Early pannus of granulation tissue growing over the patella. (e) Histology of pannus showing clear erosion of bone and cartilage at the cellular margin. (f) Histology of the pannus stained for macrophage nonspecific esterase; note long, stained dendritic processes. (g) Chronic inflammatory cells in the deeper layers of the synovium in RA. (h) A hypervillous synovium revealing well-formed secondary follicles with germinal centers (relatively rare occurrence). (i) A high power view of an area of diseased synovium showing collections of classical plasma cells. (j) Plasma cells isolated from a patient's synovial tis-

sue stained simultaneously for IgM (with fluorescein-labeled F(ab')₂ anti- μ) and rheumatoid factor (with rhodamine-labeled aggregated Fc γ). Two of the four IgM-positive plasma cells appear to be synthesizing rheumatoid factors. (k) Rheumatoid synovium showing large numbers of cells stained by anti-HLA-DR (anti-class II). (l) Rheumatoid synovium showing class II-positive accessory cells (green) in intimate contact with CD4⁺ T-cells (orange). (m) Large rheumatoid nodules on the forearm. (n) Granulomatous appearance of the rheumatoid nodule with central necrotic area surrounded by epithelioid cells, macrophages and scattered lymphocytes. Plasma cells making rheumatoid factor are often demonstrable and the lesion probably represents a response to the formation of insoluble anti-IgG complexes. (Kindly given by (a) Dr D. Isenberg; (c), (d), (e), (g), (h) and (i) Dr L.E. Glynn; (f) Professor J. Edwards; (j) Professors P. Youinou and P. Lydyard; and (k) and (l) Professor G. Janossy.)

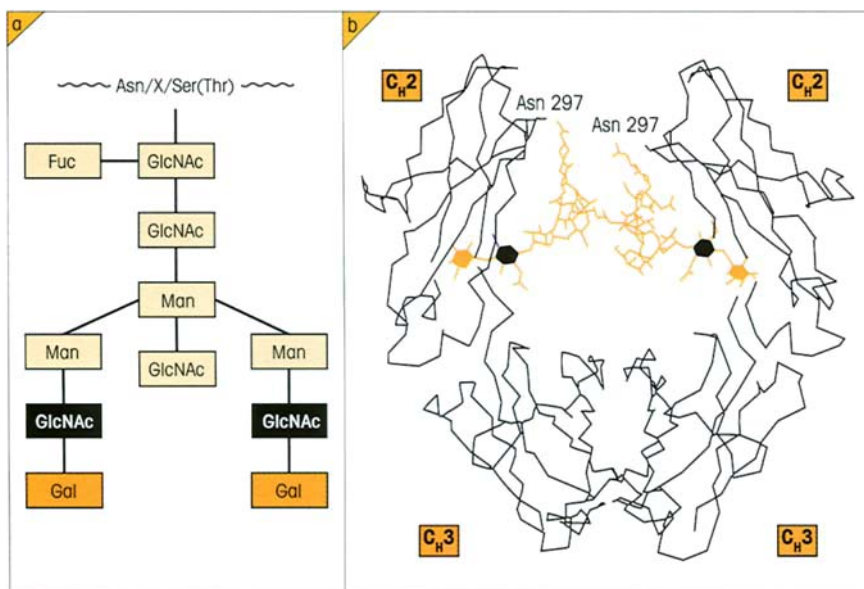


Figure 20.11. The Fc sugars and their role in bridging the two C_H2 domains of IgG. (a) Typical structure of each N-linked sugar. Some chains lack one (G1) or both terminal galactoses (agalacto-IgG; G0). (b) Structure of the C_H2 regions and the association between the terminal galactose on the 1,6 arm and the protein surface. The 1,3 arms, one of which must lack galactose, bridge the two domains. GlcNAc, N-acetylglucosamine; Man, mannose; Gal, galactose; Fuc, fucose.

- 1 The Fc may have increased autoantigenicity.
- 2 Self-associated IgG complexes (figure 20.10b) would be held together more strongly if the terminal galactose on the Fab sugar of one IgG fits into the lectin site on C_H2 left vacant by the lack of galactose on the Fc sugar.
- 3 The interaction with inflammatory mediators may be enhanced (figure 20.12). The exposure of N-acetylglucosamine in the agalacto-IgG glycoform allows recognition of immune complexes by mannose-binding protein with activation of the classical complement pathway (cf. p. 17) and stimulation of macrophages through binding of TNF. This increased inflammatory potency is clearly seen in the superiority of agalacto-anticollagen over its normally glycosylated counterpart in the production of collagen arthritis through synergy with cell-mediated immunity (see figure 20.13).
- 4 The interaction with Fc γ receptors on certain effector cells may be considerably modified as, for example, in defective feedback control of autoantibody production.

It is well established that pregnant women with RA have a remission of their disease as they approach term, but an exacerbation postpartum; as the arthritis remits, the agalacto-IgG values fall but, as the disease worsens after birth, agalacto-IgG becomes abnormal again suggesting intimate involvement with the disease process. Long-term studies in closed communities of Pima Indians, who have an unusually high incidence of RA, have shown that changes in IgG galactose provide an early marker of future clinical

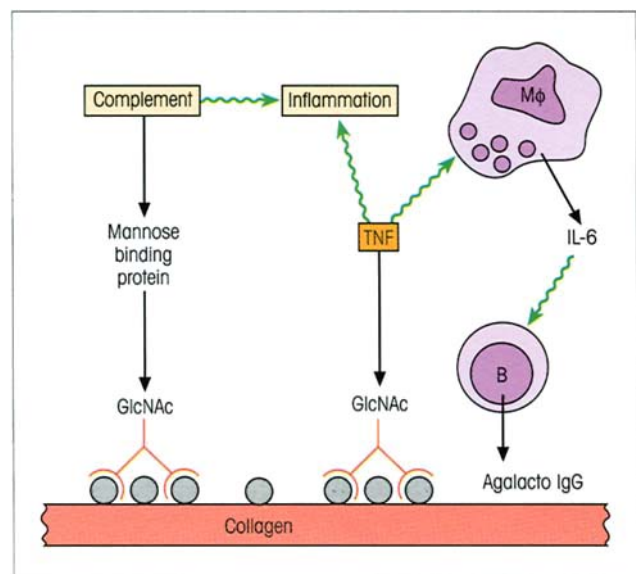


Figure 20.12. Enhancement of inflammation by complexes containing agalacto-IgG. Mannose-binding protein combines with the exposed N-acetylglucosamine (GlcNAc) and through MASP (mannose-binding protein-associated serine protease) (cf. p. 17) activates classical complement pathway C2. TNF is a lectin which also binds GlcNAc and so stimulates macrophages at the surface of the complex. The IL-6 released further promotes agalacto-IgG synthesis by B-cells.

disease and we know they can be of **prognostic value** in patients with early RA.

The production of tissue damage

As explained in the legend to figure 20.10, the complexes can be stabilized by the multivalent Fc γ -binding

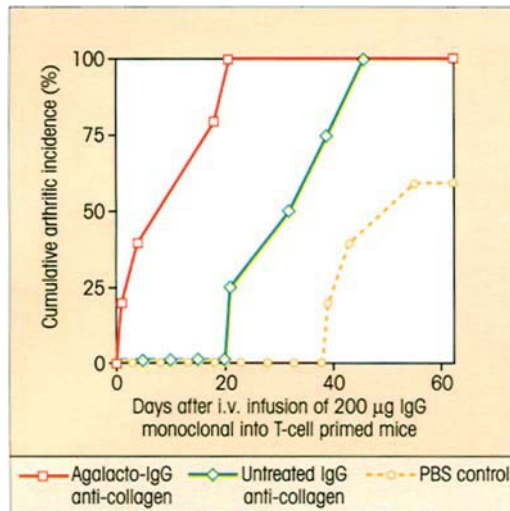


Figure 20.13. Induction of chronic arthritis by passive transfer of agalacto-IgG. Two injections of heat-denatured collagen type II in CFA (complete Freund's adjuvant) induce delayed sensitivity but no antibody or arthritis. Subsequent i.v. injection of murine monoclonal IgG anti-native collagen produced a moderate arthritis appearing after 20 days, whereas agalacto-IgG obtained by treatment of the IgG with β -galactosidase very rapidly produced a more severe arthritis. PBS, phosphate-buffered saline. (From Rademacher T.W., Williams P.J. & Dwek R.A. (1994) *Proceedings of the National Academy of Sciences of the United States of America* **91**, 6123, with permission of the National Academy of Sciences, Washington DC.)

molecules, IgM rheumatoid factor and C1q, and when present in the joint space they may initiate an Arthus reaction leading to an influx of polymorphs with which they react to release reactive oxygen intermediates (ROIs) and lysosomal enzymes. These include neutral proteinases and collagenase which can **damage the articular cartilage** by breaking down proteoglycans and collagen fibrils. More damage results if the complexes are adherent to the cartilage, since the polymorph binds but is unable to internalize them ('frustrated phagocytosis'); as a result, the lysosomal hydrolases are released extracellularly into the space between the cell and the cartilage where they are protected from enzyme inhibitors such as α_2 -macroglobulin. We have already drawn attention to the enhanced inflammatory potency of complexes containing agalacto-IgG.

The aggregates may also stimulate the macrophage-like cells of the synovial lining, either directly through their surface receptors or indirectly through phagocytosis and resistance to intracellular digestion. At this point we should acknowledge that the release of cytokines such as TNF and GM-CSF from activated T-cells provides further potent macrophage stimulation.

The activated synovial cells grow out as a malign pannus (cover) over the cartilage (figure 20.8d) and, at

the margin of this advancing granulation tissue, breakdown can be seen (figure 20.8e), almost certainly as a result of the release of enzymes, ROIs and especially of IL-1, 6 and TNF. Activated macrophages also secrete plasminogen activator and the plasmin formed as a consequence activates a latent collagenase produced by synovial cells. Sensitization to partially degraded collagen may occur and this could lead secondarily to amplification of the lesion. The secreted products of the stimulated macrophage can activate chondrocytes to exacerbate **cartilage breakdown**, and osteoclasts to bring about **bone resorption** which is a further complication of severe disease (figure 20.8c). Subcutaneous nodules are granulomas (figure 20.8m and n) possibly formed through local production of insolubilized self-associating antiglobulins.

The contribution of these immune complexes to the pathogenesis of RA is built into the overview presented in figure 20.14, where it will be seen that a role for the T-cells must by no means be overlooked and will be discussed in the following section.

T-CELL-MEDIATED HYPERSENSITIVITY AS A PATHOGENIC FACTOR IN AUTOIMMUNE DISEASE

Rheumatoid arthritis again

The chronically inflamed synovium is densely crowded with activated T-cells and their critical role in the disease process is emphasized by the beneficial effects of cyclosporin and anti-CD4 treatments, and by the increased risk of disease associated with the 'shared epitope' sequences Q(R)K(R)RAA from residues 70–74 on the DR β chain of DR1 and certain DR4 alleles (cf. table 17.2). High levels of IL-15 within the synovial membrane can recruit and activate T-cells whose secretion of cytokines and ability to induce macrophage synthesis of TNF and further IL-15 will drive pannus development powerfully with consequent erosion of cartilage and bone (figure 20.14). Chondrocytes themselves may also be disease targets.

Just as in SLE, the antigenic specificity of these T-cells is still unknown. An appealing clue has come from the finding that the QKRAA shared epitope sequence, which lies within a polymorphic region of HLA-DR4/1 subtypes, is also present in the dnaJ heat-shock proteins from *E. coli*, *Lactobacillus lactis* and *Brucella ovis*, as well as the Epstein-Barr virus gp110 protein. This already provides an opportunity for priming of T-cells with autoreactive specificity for a processed peptide containing QKRAA presented by another HLA molecule, as discussed previously

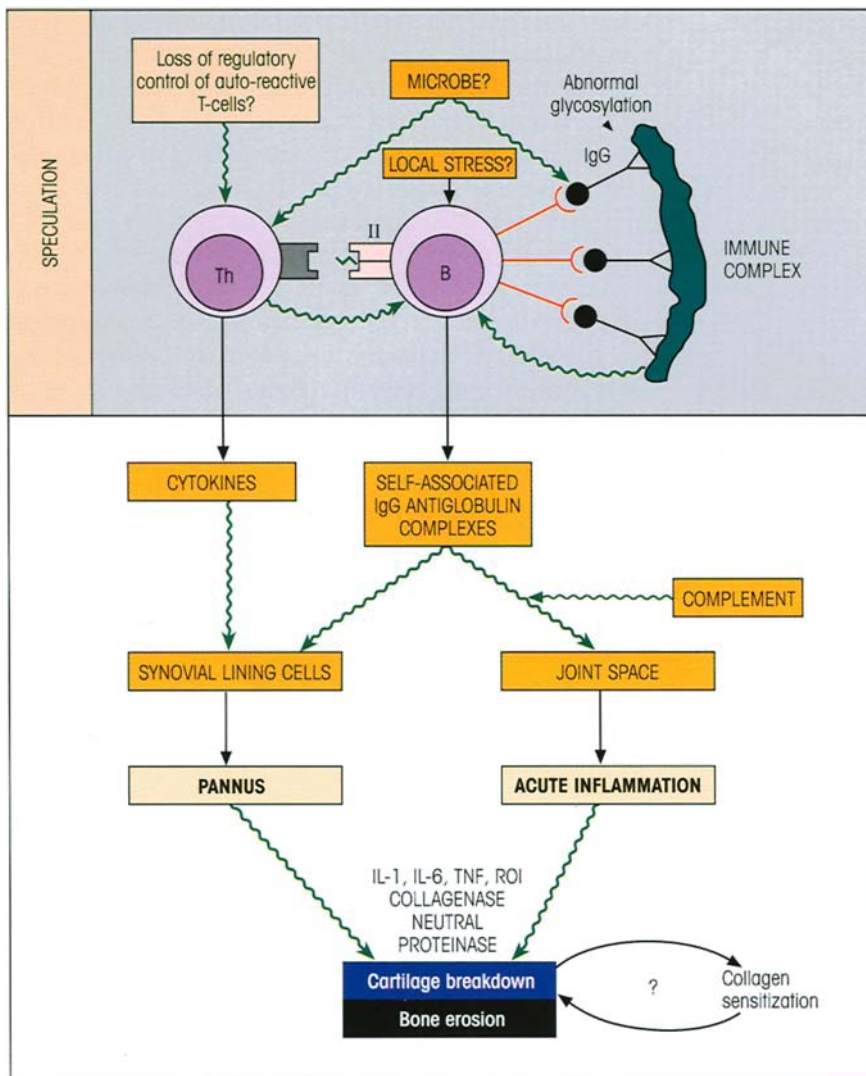


Figure 20.14. Immune pathogenesis of rheumatoid arthritis and speculation on the induction of autoimmunity. The defective pituitary–adrenal regulatory control on cell-mediated immunity in these patients was described in Chapter 19. The identity of the peptide(s) associated with the class II molecule is unknown. There are numerous opportunities for cross-reactions between the DR β ‘shared epitope’ and various microbial heat-shock proteins. It has been reported that synovial fluid T-cells of juvenile rheumatoid arthritis patients respond strongly to human heat-shock protein 60, whereas those of adult rheumatoid arthritis patients do not. Recent data (Matsumoto *et al.* (1999) *Science* **286**, 1732) accord with the view that there must be some feature of the joint which makes it particularly sensitive to immune responses, maybe especially immune complexes, involving ubiquitous antigens.

(p. 414). The plot deepens with the realization that QKRAA binds to a second *E. coli* heat-shock protein dnaK and that HLA-DR containing the QKRAA sequence binds the *self* analogue of dnaK, namely hsp73, which targets selected proteins to lysosomes for processing. What this all means remains to be resolved but note the involvement of the hsp family yet again. Suspicion of previous microbial encounters is engendered by the discovery by PCR amplification of nucleotide sequences characteristic of *Mycoplasma fermentans*, *Chlamydia* and Epstein–Barr virus in a substantial proportion of synovial tissues removed from RA patients. Further work on the intracellular location, molecular status and possible expression of this material is awaited with interest.

The antigenic history of **reactive arthritis** is more amenable to study since it is triggered by an infection either of the urogenital tract by *Chlamydia trachomatis*

or of the enteric tract with *Yersinia*, *Salmonella*, *Shigella* or *Campylobacter*. The synovial tissue in reactive arthritis remarkably still retains antigenic descendants or memorials of the initiating bacteria many years after infection which can drive local T-cells. All the microbes are either obligate or facultative intracellular bacteria and so may escape the immune system by hiding inside cells, probably aided by high local production of IL-4. However, we may be dealing with molecular mimicry. Natural infection with *Salmonella typhimurium* generates CD8 cytotoxic T-cells which recognize an immunodominant epitope of the GroEL molecule presented by the class Ib Qa-1 and cross-react with a peptide from mouse hsp60, so permitting a reaction with stressed macrophages. HLA-B27 individuals are particularly at risk and the importance of the microbial component is emphasized by experiments on mice bearing the B27 transgene; if reared in a germ-free

environment, lesions are restricted to the skin, but in the microbiological wilderness of the normal animal house, the skin, gut and joints are all affected. Why, as in RA, are the joints targeted and what does B27 do? Only one in 300 of the T-cells in the reactive arthritis synovium is CD8 and therefore class I restricted. It could be that a cross-reactive B27 sequence functions as a cryptic epitope perpetuating a gentle microbial stimulus with an amplifying autoimmune response.

Two experimentally induced models of arthritis are heavily dependent on T-cells. **Adjuvant arthritis** resulting from immunization of rats with complete Freund's adjuvant (CFA) only can be transferred to naive recipients with a T-cell clone specific for the mycobacterial heat-shock protein hsp60. The **collagen arthritis** model involves injection of type II collagen in complete Freund's, but here a synergy between cell-mediated hypersensitivity and antibody seems to operate: sensitization with denatured collagen in CFA-induced T-cell-mediated immunity but no arthritis unless the mice were also given IgG from animals primed with *native* collagen (figure 20.13).

Spontaneous models of arthritis are hard to come by. Up to 30% of MRL/*lpr* mice have arthritic lesions but this incidence varies with the animal house. If animals harbor Sendai virus, they are protected from arthritis; what is that telling us? To complete the circle, they also have raised levels of agalacto-IgG.

Before leaving the subject, one should not overlook the curious behavior of fibroblasts isolated from synovial tissue in established rheumatoid arthritis. In culture, they secrete collagenase which would contribute to cartilage breakdown, but more ominously they proliferate spontaneously due to unrestrained control of the cell cycle by cyclin-dependent kinases 4 and 6 and their respective D-cyclins. This implies that they would be refractory to conventional therapies and might be a cause of apparently intractable disease.

Organ-specific endocrine disease

To make a fairly sweeping statement, inflammatory organ-specific diseases are generally linked to T-helper-1 (Th1) responses. Clones producing EAE or transferring diabetes from NOD mice produce IL-2 and γ -interferon (IFN γ), while in collagen arthritis IL-12 can be substituted for the mycobacteria in the complete Freund's adjuvant. On the other hand, Th2 CD4s are responsible for the polyclonal activation in murine lupus, the glomerulonephritis and necrotizing vasculitis induced in Brown Norway rats by mercuric chloride, and the chronic autoimmunity generated during graft-vs-host disease. We will see that Th2 re-

sponses can downregulate destructive Th1 cells. Last, the Th1/Th2 polarization is not apparent in diseases such as myasthenia gravis, Graves' thyrotoxicosis, Sjögren's syndrome and primary biliary cirrhosis.

Autoimmune thyroiditis

The inflammatory infiltrate in autoimmune thyroiditis is usually essentially mononuclear in character (see figure M19.1.1c) and, although not an infallible guide, this has been taken as an expression of T-cell-mediated hypersensitivity. Firm evidence for a direct participation of T-lymphocytes has yet to be provided, although the demonstration of class II molecules on patients' thyrocytes and the presence of antigen-specific Th1 cells in the thyroid would accord with an involvement of these cells.

We must turn to the animal models for further evidence albeit indirect. Draconian stamping out of T-cells in the Obese strain chicken by neonatal thymectomy, and repeated injection of anti-T-cell serum, prevented the spontaneous development of atrophic autoimmune thyroiditis. It is also of interest to note that, at the target cell level, the threshold for induction of MHC class II on OS thyrocytes by IFN γ is far lower than that reported for normal thyroid cells, further reinforcing the notion that a thyroid abnormality is a contributory factor to the susceptibility phenotype. The other model, in which thyroiditis is induced by thyroglobulin in complete Freund's adjuvant (see figure M19.1.1b), can be transferred to naive histocompatible recipients with CD4⁺ T-cell clones specific for peptides containing thyroxine established from immunized animals. The cells infiltrate between the thyroid follicles and probably kill the epithelial cells by a combination of locally released IFN γ and TNF. We see now that there is considerable diversity in the autoimmune response to the thyroid leading to tissue destruction, metabolic stimulation, growth promotion or mitotic inhibition which in different combinations account for the variety of forms in which autoimmune thyroid disease presents (figure 20.15).

Insulin-dependent diabetes mellitus (IDDM)

Just as in autoimmune thyroiditis, IDDM involves chronic inflammatory infiltration and destruction of the specific tissue, in this case the insulin-producing β -cells of the pancreatic islets of Langerhans. The delay in onset of disease achieved by early treatment with cyclosporin, at levels which have little effect on antibody production, points an accusing finger at effector T-cells as the agents of destruction, since this drug tar-

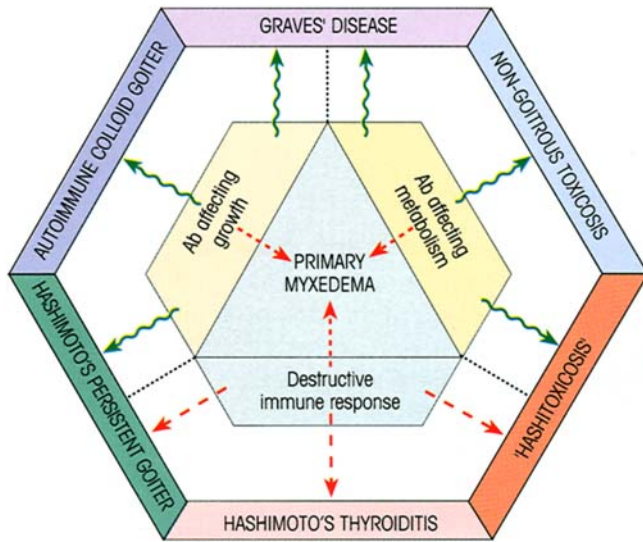


Figure 20.15. Relationship of different autoallergic responses to the circular spectrum of autoimmune thyroid diseases. Responses involving thyroglobulin and the thyroid peroxidase (microsomal) surface microvillous antigen lead to tissue destruction, whereas autoantibodies to TSH (and other ?) receptors can stimulate or block metabolic activity or thyroid cell division. 'Hashitoxicosis' is the down-to-earth term used by our Scots colleagues to describe a gland showing Hashimoto's thyroiditis and thyrotoxicosis simultaneously. (Courtesy of Professors D. Doniach and G.F. Bottazzo.)

gets T-cell cytokine synthesis so specifically. *In vitro* T-cell responses to islet cell antigens including glutamic acid decarboxylase (GAD) directly reflect the risk of progression to clinical IDDM. The strength of the risk factors associated with certain HLA-DQ alleles also has a strong whiff of T-cell action although, quite mysteriously, monocytes (and dendritic APCs?) constitutively expressing cyclooxygenase-2 (COX-2), the inducible enzyme responsible for the synthesis of prostaglandin E₂ and other prostanoids, were present in relatives (of IDDM patients) bearing these alleles and in the patients themselves.

To obtain further insight into the cellular siege and destruction of the islet β -cells, one has to turn to the **Nonobese diabetic (NOD) mouse** which spontaneously develops diabetic disease closely resembling human IDDM in its range of autoimmune responses and the association of islet breakdown with a chronic infiltration by T-cells and macrophages (figure 20.16). T-cells infiltrating the islets in diabetic mice had a Th1-type cytokine profile and could transfer disease to NOD recipients congenic for the severe combined immunodeficiency (*SCID*) mutation. The CD4/CD45RB^{lo} memory subset of splenocytes from

young nondiabetic NOD mice inhibited this transfer, but the same subset from older diabetic mice had lost its regulatory power and had become pathogenic. This change to a Th1-type response coincided with the onset of diabetes and is consistent with the observation that IL-12, which favors Th1 responses, exacerbates disease. Just as in the human disease, MHC class II alleles hold a pivotal controlling position and introduction of a transgene, in which residues at position 56 or 57 of the H-2A β chain are altered, drastically inhibits the development of diabetes. One of the non-MHC susceptibility loci in NOD mapped to the *IL-1R* and *Bcg* genes on chromosome 1 was associated with natural resistance to infection by intracellular parasites. As a result, the NOD mouse is resistant to *Mycobacterium avium*, but after recovery from infection the onset of diabetes is prevented. That responses to hsp60 may be involved is implied by reports that a 24-amino acid peptide from this protein is the target of diabetogenic T-cells in both patients and the NOD mouse, and treatment with this peptide or with a NOD peptide-specific Th1 clone downregulates spontaneous disease. Cohen has interpreted this as a result of dysregulation of the 'immunological homunculus' (cf. p. 208) and notes that the level of a particular idio type associated with a TCR CDR3 consistently falls prior to the onset of diabetes, while mice reared under germ-free conditions which might inhibit the development of a natural idio type network are more susceptible to IDDM. Time will test the validity of this hypothesis. Notwithstanding this evidence relating to hsp and idio types, in the final analysis, one has to take into account the following: up to 50% of the infiltrating T-cells isolated from pre-diabetic NOD islets are insulin-specific and can transfer disease to young NOD mice; GAD-specific T-cells can also be recovered and are also diabetogenic; and tolerance to either insulin or GAD prevents the onset of disease. Presumably the latter can be accommodated by an organ-related bystander tolerance mechanism described below (p. 445). Overall, the data seem to be consistent with the necessity for two pathogenic pathways, one dependent on hsp and the other on an organ-specific response, operating either synergistically or serially, to achieve the final destruction of the pancreatic β -cells.

GAD in the central and peripheral nervous system produces γ -aminobutyric acid (GABA), a major inhibitory neurotransmitter, from glutamine. Autoantibodies to GAD are seen not only in early diabetes, but also in **Stiff man syndrome** (sounds like a cue for a Western) where the GABA-ergic pathways controlling motor neuron activity are defective. The antibodies cannot be pathogenic because GAD is present on the

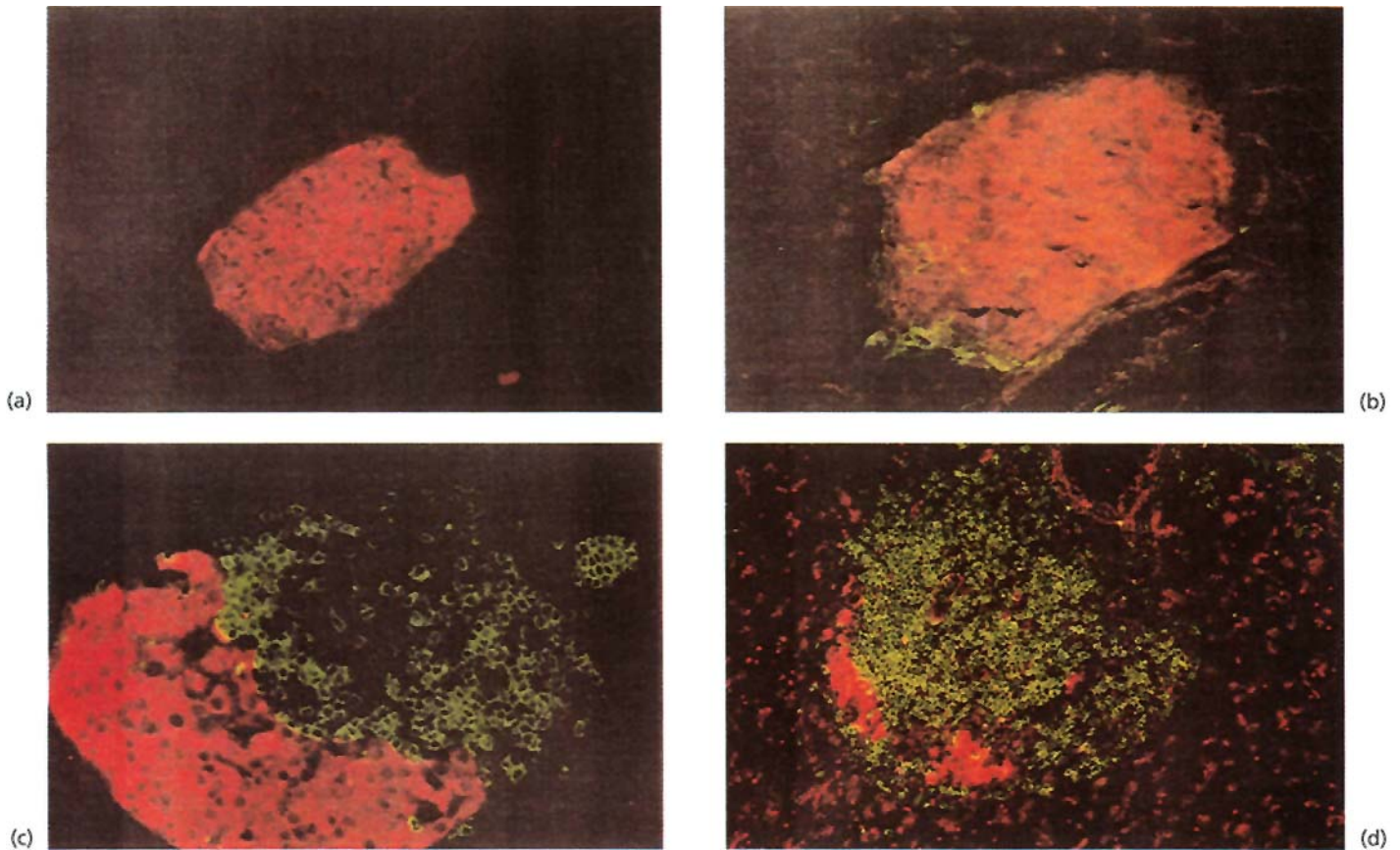


Figure 20.16. Destruction of pancreatic islet β -cells by infiltrating T-cells in the Nonobese diabetic (NOD) mouse. (a) Normal intact islet. (b) Early peri-islet infiltration. (c) Penetration of the islet by infiltrating T-cells. (d) Almost complete destruction of insulin-producing cells with replacement by invading T-cells. Insulin

stained by rhodamine-conjugated antibodies and T-cells by fluoresceinated anti-CD3. (Data reproduced from Quartey-Papafio R., Lund T., Cooke A. *et al.* (1995) *Journal of Immunology* **154**, 5567; photographs kindly provided by Dr Jenny Phillips.)

inner surface of the plasma membrane, but T-cells could be. How the brain as distinct from the pancreatic islet could be specifically targeted is a conundrum but 30% of patients do develop IDDM.

Multiple sclerosis (MS)

The idea that MS could be an autoimmune disease has for long been predicated on the morphological resemblance to *experimental autoimmune encephalomyelitis* (EAE), a demyelinating disease leading to motor paralysis (figure 20.17) produced by immunization with myelin, usually *myelin basic protein* (MBP) in complete Freund's. T-cell clones specific for MBP belong to restricted TCR V β families. They will transfer disease but this can be exacerbated by injection of a monoclonal antibody to Theiler's virus, a murine encephalomyelitis virus, cross-reacting with an epitope on myelin and oligodendrocytes. Presum-

ably the T-cell incites a local inflammation affecting the endothelial cells at the blood–brain barrier which opens the gate for antibody to penetrate the brain tissue.

How much of this is relevant to human disease? First, the serologically determined Caucasian DR2 phenotype (DRB1*1501, DQA1*0102, DQB1*0602) is strongly associated with susceptibility to MS. At least 37% of activated T-cells responsive to IL-2/4 in cerebrospinal fluid were specific for myelin components, compared with a figure of 5% for subjects with other neurological disturbances. A Leu.Arg.Gly. amino acid sequence motif found in around 40% of TCR V β 5.2 N(D)N rearrangements in T-cells from MS lesions was present in a V β 5.2 clone from an MS patient cytotoxic towards targets containing the MBP 89–106 peptide and in encephalitogenic rat T-cells specific for MBP peptide 87–99. One is greatly encouraged to continue with attempts to induce tolerance.

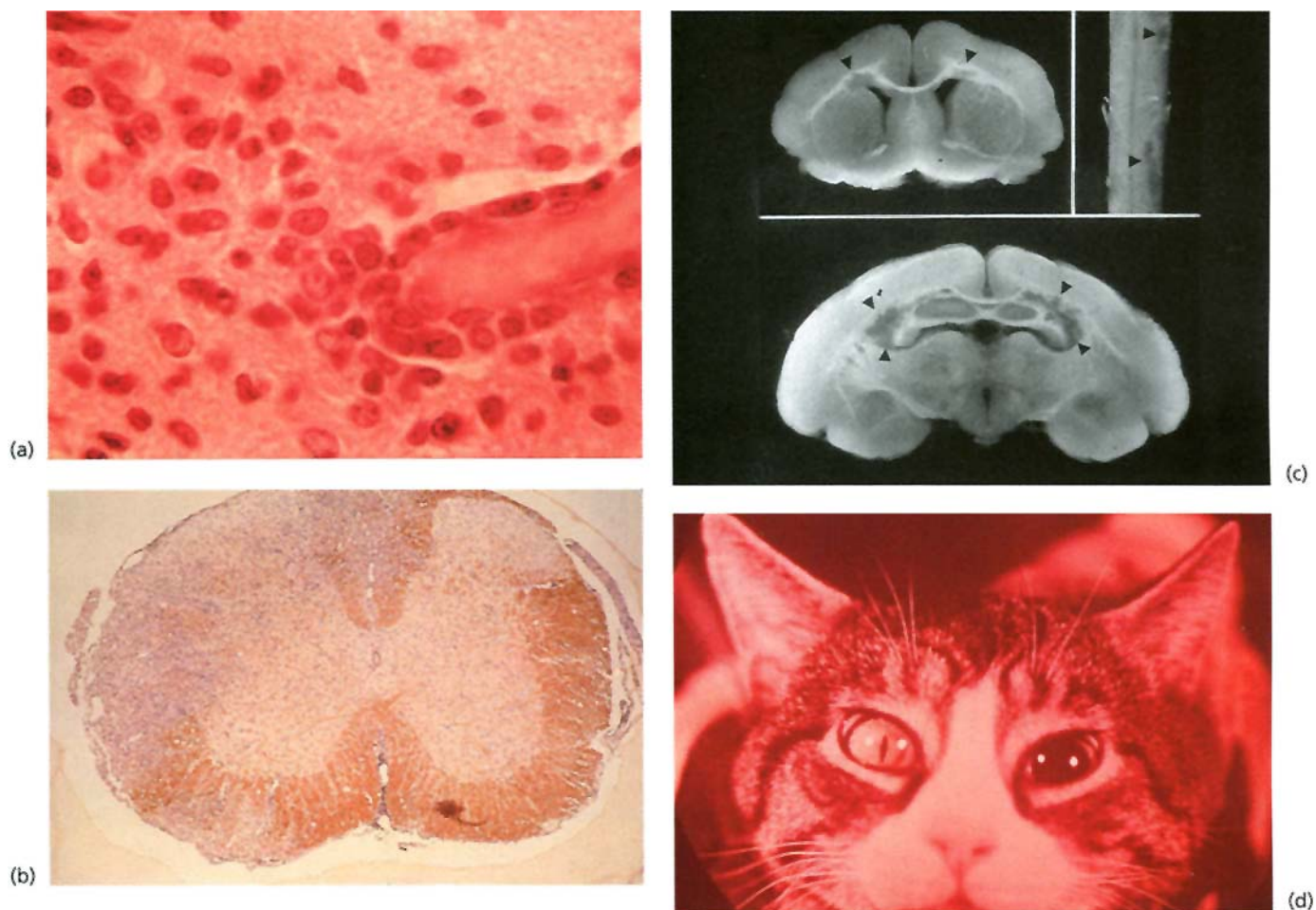


Figure 20.17. Experimental autoimmune encephalomyelitis (EAE), a demyelinating model for multiple sclerosis induced by immunization with brain antigens in complete Freund's adjuvant (CFA). (a) Early lesion of EAE in the rat at 9 days after immunization with rat spinal cord homogenate in CFA. The lesion in brain white matter, which is probably a few hours old, shows perivenous infiltration of lymphocytes and monocytes (a pure mononuclear inflammation) with cells invading the nervous parenchyma. Myelin is not stained. (b) Lumbar spinal cord of rat with chronic EAE after immunization with myelin proteolipid protein. Large demyelinating

lesions in dorsal columns, in both left (large) and right (small) columns, as well as on lower left. Also gray matter involved with on-going inflammation, in particular affecting left dorsal horn. Normal myelin is stained brown. (c) Chronic relapsing EAE in guinea-pig. Large demyelinated plaques in brain white matter (arrows) closely similar to plaques of multiple sclerosis. (d) Acute EAE in cat with optic nerve involvement. (Legend and slides provided by Dr B. Waksman; (b) originally from Dr Trotter; (c) from Drs Lassmann and Wisniewski and (d) from Dr Patterson.)

Psoriasis

Given the evidence for T-cell-mediated pathogenesis (p. 344), the isolation of clones specific for group A β -hemolytic streptococci from guttate skin lesions has fostered the thought that pathology is initiated by exotoxin (i.e. superantigen) recruited T-cells and is maintained by specific cells reacting both with streptococcal M protein and a cryptic skin epitope, possibly a keratin variant presented by cytokine-activated keratinocytes. There is extensive sequence homology between M proteins and type I keratin.

SOME OTHER SYSTEMIC VASCULAR DISORDERS WITH IMMUNOPATHOLOGICAL COMPONENTS

The characteristic feature of **Wegener's granulomatosis** is a necrotizing granulomatous vasculitis. Although the antibodies to proteinase III (antineutrophil cytoplasmic antibodies (cANCA); figure 20.18 and table 19.2) characteristic of this disease are directed to an intracellular antigen associated with the primary granules of the polymorph, recent studies reveal a possible mechanism by which they might induce

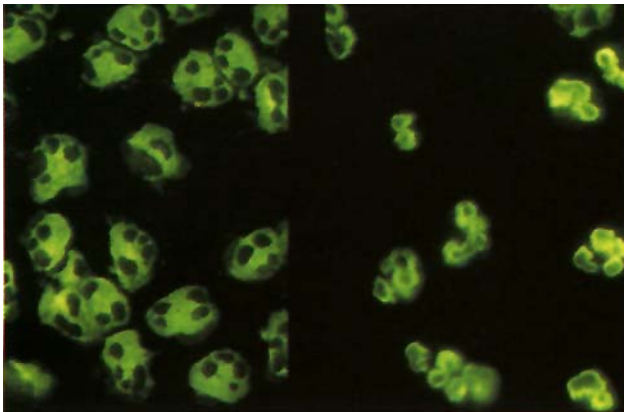


Figure 20.18. Antineutrophil cytoplasmic antibodies (ANCA). *Left*—cytoplasmic cANCA (cf. table 12.2) diffuse staining specific for proteinase III in Wegener's granulomatosis; *right*—perinuclear p-ANCA staining by myeloperoxidase antibodies in periarthritis nodosa. Fixed neutrophils are treated first with patient's serum then fluorescein-conjugated anti-human Ig. (Kindly provided by Dr G. Cambridge.)

vasculitic lesions. Cytokine priming of polymorphs causes translocation of proteinase III to the cell surface, whereupon reaction with the autoantibody activates the cell causing degranulation and generation of reactive oxygen intermediates (ROIs). A possible scenario might go something like this: tumor necrosis factor (TNF) induced by infection could activate endothelial cells to secrete interleukins IL-1 and IL-8 which attract neutrophils, upregulate their lymphocyte function associated molecule-1 (LFA-1) adhesion molecules, and prime them for reaction with the proteinase III antibody. Endothelial cell injury would then be a consequence of the release of superoxide anion and other ROIs. Other authors have laid claim to endothelial membrane antibodies which upregulate adhesion molecules and increase secretion of IL-6 and 8 and MCP-1. Anti-ANCA idiotype is demonstrable in the IgM fraction of serum from patients in remission and their removal uncovers underlying IgG ANCA activity.

A giant cell arteritis of large- and medium-sized arteries, involving mainly CD8⁺ T-cells and macrophages, characterizes the pathology of **temporal arteritis**. The antigen is elusive but the disease is strongly associated with HLA-DR4 and is exquisitely sensitive to high-dose steroids.

Systemic sclerosis, also known as **scleroderma**, is a generalized disease of connective tissue with increased deposition of collagen and other matrix components, causing extensive fibrotic destruction of the skin and internal organs centered around small arteries and microvasculature, eventually producing

capillary occlusion. To put it mildly, the pathogenesis is poorly understood, but the high frequency of positive results for centromere, nucleolar and topoisomerase-1 (Scl-70) autoantibodies and rheumatoid factors betokens some major intrusion by autoimmune elements, and there is multisystem infiltration by CD8⁺ T-cells secreting mainly TGFβ and IL-6. So far, the only clues pointing to early etiological factors have come from animal models of scleroderma. So-called 'tight skin' mice have a mutation in the fibrillin-1 gene coding for an extracellular matrix protein which might be responsible for activating T-cells. Another model, the UCD 200 chicken, reveals endothelial cell apoptosis as a very early event, speculatively due to ADCC mediated by endothelial cell antibodies; mononuclear cell infiltration occurred later with fibrosis as the hallmark of late disease. Anyway, in human scleroderma, TGFβ secreted by T-cells would stimulate dermal fibroblasts to overproduce collagen and, ultimately, just like the synovial fibroblasts in RA, they acquire a semi-autonomy making the disease very difficult to treat.

Atherosclerotic plaques are focal lesions of large elastic and muscular arteries producing intimal thickening and are composed of a subendothelial fibrous cap of collagen and matrix-rich connective tissue, lipid-filled macrophages, proliferating smooth muscle cells and some CD4 T-lymphocytes. Rupture of a plaque leads to thrombosis. The mood is swinging towards the idea that autoimmunity may initiate or exacerbate the process of plasma lipid deposition and plaque formation. The two lead candidate antigens are heat-shock protein 60 (hsp60) and the low density lipoprotein (LDL), apoprotein B, which is the main carrier of cholesterol.

The indicative evidence is as follows (figure 20.19). Immunization with mycobacterial hsp65 elicits atherosclerotic lesions at classical predilection sites subject to major hemodynamic stress, and a cholesterol-rich diet makes them worse. Antibodies are produced which are said to react with heat- or TNF-stressed endothelial cells, implying that hsps are somehow implicated. Furthermore, such endothelial provocation and activation of the macrophages by contact with LDL generate oxidative free radicals, which convert lecithin within the LDL to lysolecithin which is chemoattractant and cytotoxic, and polyunsaturated fatty acids to alkenals which react with the lysine residues of apoprotein B making it 'tasty' for the macrophage scavenger receptors and, perhaps, auto-antigenic. Pointing the finger at these oxidative processes as keys to atherogenesis receives support from the epidemiological data on an inverse correlation between coronary heart disease in patients and

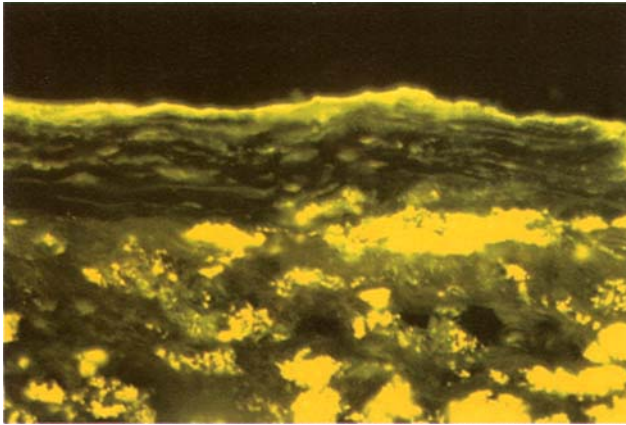


Figure 20.19. Expression of heat-shock protein 60 in an early human arteriosclerotic lesion. Frozen, unfixed, 4 μ m thick section of a fatty streak (=early lesion) of a human carotid artery stained in indirect immunofluorescence with a monoclonal antibody to heat-shock protein 60 and a fluorescein-labeled second antibody. A strong reaction with endothelial cells as well as cells infiltrating the intima, including foam cells, is evident. (Original magnification $\times 400$.) (Photograph kindly provided by Professor G. Wick.)

dietary intake of antioxidants, and the observation that the induction of atherosclerotic fatty streaks by high cholesterol diets in rabbits can be prevented by the antioxidant probucol, despite its lack of influence on plasma cholesterol levels. Yet another piece in the jigsaw: β_2 -glycoprotein-1, which figures prominently in the antiphospholipid syndrome (cf. p. 421), is also abundantly present in atherosclerotic lesions. Clarification is awaited.

DIAGNOSTIC VALUE OF AUTOANTIBODY TESTS

Serum autoantibodies frequently provide valuable diagnostic markers. The most useful routine test is screening of the serum by immunofluorescence on a frozen section prepared from a composite block of unfixed human thyroid and stomach, and rat kidney and liver. This is supplemented by agglutination tests for rheumatoid factors and for thyroglobulin, thyroid peroxidase and red cell antibodies and by ELISA for antibodies to intrinsic factor, DNA, IgG, extractable nuclear antigens, and so on (see table 19.2). The salient information is summarized in table 20.1. ELISAs are taking over and tests with purified gene-cloned antigens arranged in minispot arrays will one day supplant the need for immunofluorescence which is time-consuming and more skilled.

The tests will also prove of value in screening for people at risk, e.g. relatives of patients with autoimmune diseases such as diabetes, thyroiditis patients for

gastric autoimmunity and vice versa, and ultimately the general population if the sociological consequences are fully understood and acceptable.

The development of easily performed commercial ELISPOT (cf. p. 140) assay kits for monitoring the production of individual cytokines by peripheral blood T-cells incubated with specific antigens will be greatly welcomed.

TREATMENT OF AUTOIMMUNE DISORDERS

Control at the target organ level

The majority of approaches to treatment, not unnaturally, involve manipulation of immunological responses (figure 20.20). However, in many organ-specific diseases, metabolic control is usually sufficient, e.g. thyroxine replacement in primary myxedema, insulin in juvenile diabetes, vitamin B₁₂ in pernicious anemia, antithyroid drugs for Graves' disease, and so forth. Anticholinesterase drugs are commonly used for long-term therapy in myasthenia gravis; thymectomy is of benefit in most cases and it is conceivable that the gland contains acetylcholine (ACh) receptors in a particularly immunogenic form (? associated with HLA class II expression).

It is worth recording that maintenance therapy to replace the loss of an organ-specific molecule, such as insulin in IDDM, might have the effect of subduing metabolic activity and reducing expression of the target antigen. Help should be on the way for patients with burnt-out pancreatic β -cells. Xenografts of genetically engineered fetal or neonatal pig islets (cf. figure 17.15) are under study, and other good news is that stem cells contained within ductal structures of the adult pancreas can be differentiated in culture to provide a 10 000-fold increase in the number of available islets per organ, although any recurrence of immunologically mediated damage would need to be dealt with. Perhaps transfection of the grafted cells with TGF β might provide an appropriately suppressive microenvironment. A slightly cheeky but novel approach to the repair of damaged target organs is to target a nonpathogenic specific T-cell clone expressing a transgenic growth factor to the inflamed area. Thus, a nonaggressive Th2 clone, specific for the proteolipid (PLP) brain antigen, delivered a platelet-derived growth factor (PDGF-A) transgene to the inflamed brain of an animal with EAE; contact with antigen stimulated the clone and the secreted growth factor induced proliferation of the oligodendrocyte progenitor cells involved in remyelination. This would be a highly customized therapy only suitable for the well-

Table 20.1. Autoimmunity tests and diagnosis.

DISEASE	ANTIBODY	COMMENT
Hashimoto's thyroiditis	Thyroid	Distinction from colloid goiter, thyroid cancer and subacute thyroiditis Thyroidectomy usually unnecessary in Hashimoto goiter
Primary myxedema	Thyroid	Tests +ve in 99% of cases. If suspected hypothyroidism assess 'thyroid reserve' by TRH stimulation test
Thyrotoxicosis	Thyroid	High titers of cytoplasmic Ab indicate active thyroiditis and tendency to post-operative myxedema: anti-thyroid drugs are the treatment of choice although HLA-B8 patients have high chance of relapse
Pernicious anemia	Stomach	Help in diagnosis of latent PA, in differential diagnosis of non-autoimmune megaloblastic anemia and in suspected subacute combined degeneration of the cord
Insulin-dependent diabetes mellitus (IDDM)	Pancreas	Insulin Ab early in disease. GAD Ab standard test for IDDM. Two or more autoAb seen in 80% of new onset children or prediabetic relatives but not in controls
Idiopathic adrenal atrophy	Adrenal	Distinction from tuberculous form
Myasthenia gravis	Muscle ACh receptor	When positive suggests associated thymoma (more likely if HLA-B12), positive in >80%
Pemphigus vulgaris and pemphigoid	Skin	Different fluorescent patterns in the two diseases
Autoimmune hemolytic anemia	Erythrocyte (Coombs' test)	Distinction from other forms of anemia
Sjögren's syndrome	Salivary duct cells, SS-A, SS-B	
Primary biliary cirrhosis	Mitochondrial	Distinction from other forms of obstructive jaundice where test rarely +ve Recognize subgroup within cryptogenic cirrhosis related to PBC with +ve mitochondrial Ab
Active chronic hepatitis	Smooth muscle anti-nuclear and 20% mitochondrial	Smooth muscle Ab distinguish from SLE Type 1 classical in women with Ab to nuclei, smooth muscle, actin and asialoglycoprotein receptor (these Ab disappear on remission indicating reduction in steroids) Type 2 in girls and young women with anti-LKM-1 (cyt P450)
Rheumatoid arthritis	Antiglobulin, e.g. SCAT and latex fixation Antiglobulin + raised agalacto-Ig Perinuclear	High titer indicative of bad prognosis Prognosis of rheumatoid arthritis V.sp. for early RA. Dominant residue citrulline (post-translational modification of arginine)
SLE	High titer antinuclear, DNA Phospholipid	DNA antibodies present in active phase Ab to double-stranded DNA characteristic; high affinity complement-fixing Ab give kidney damage, low affinity CNS lesions Thrombosis, recurrent fetal loss and thrombocytopenia
Scleroderma	Nucleolar + centromere Scl-70	Characteristic of the disease
Wegener's granulomatosis	Neutrophil cytoplasm	Antiserine protease closely associated with disease; treatment urgent

heeled, but a cheaper gambit is to inject a plasmid DNA coding for Fas ligand in liposomes directly into the organ under attack. This manipulation, carried out in an experimental autoimmune thyroiditis model, induced persisting expression of FasL on thyroid follicular cells and total abrogation of antithyroglobulin cytotoxic T-lymphocytes.

The reader may recall the spontaneous proliferation of RA synovial fibroblasts when placed in culture. These cells undergo irreversible cell cycle arrest if sub-

jected to γ -irradiation, which induces the synthesis of the $p16^{INK4a}$ senescence protein. This is a tumor suppressor which blocks the stable association of cyclin-dependent kinases 4 and 6 with their respective D-cyclins and inhibits their ability to mastermind the passage of cells into the G1 phase of the cell growth cycle. Injection of the RA fibroblasts with a recombinant adenovirus encoding the $p16^{INK4a}$ gene halted their growth and reduced synovial cell hyperplasia in the adjuvant arthritis model. If focus on expression of

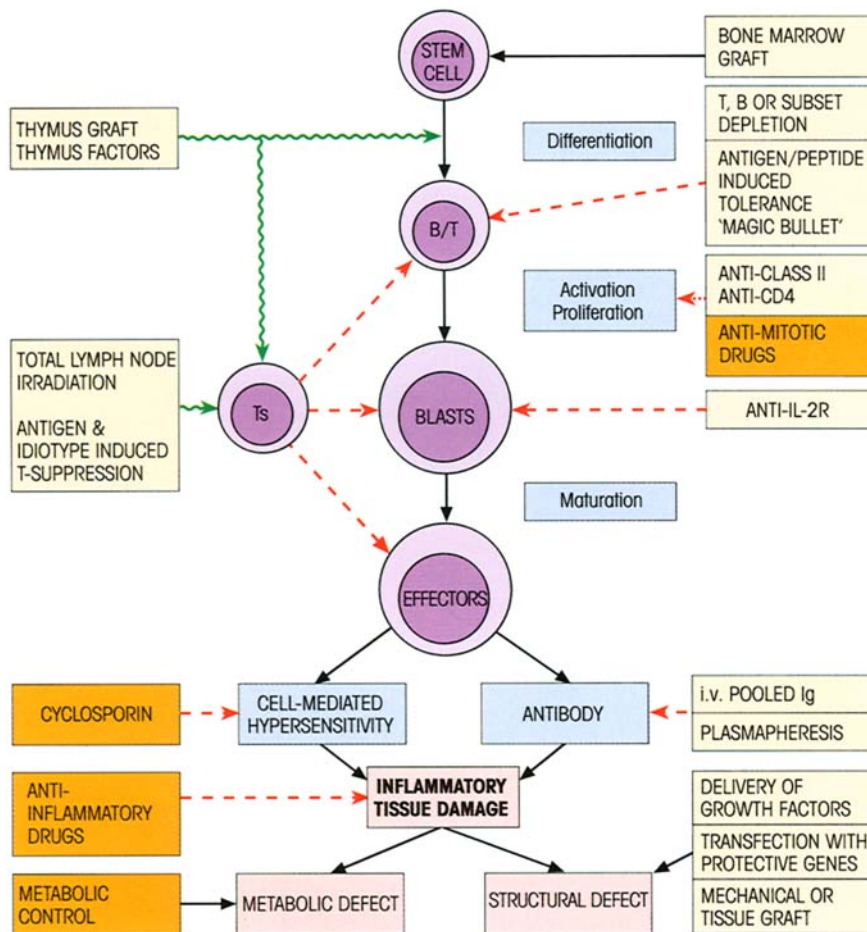


Figure 20.20. The treatment of autoimmune disease. Current conventional treatments are in dark orange; some feasible approaches are given in lighter orange boxes. (In the case of a live graft, bottom right, the immunosuppressive therapy used may protect the tissue from the autoimmune damage which affected the organ being replaced.)

this gene fosters a new therapeutic approach to RA, it might also find utility in other disorders such as atherosclerosis, scleroderma and possibly late stage asthma. Conceivably, the benefit of adding methotrexate to anti-TNF therapy in RA (see below) might be partly attributable to an effect on fibroblast proliferation.

Based on the possibility that multiple sclerosis is virally driven, patients have been treated with IFN β ; relapse rates were reduced by a third in relapsing-remitting disease, but there was only a modest effect on progressive disease. Note, however, that IFN β influences some T-cell functions in addition to its effect on viral proliferation.

Anti-inflammatory drugs

Patients with severe myasthenic symptoms respond well to high doses of steroids and the same is true for serious cases of other autoimmune disorders, such as SLE and immune complex nephritis, where the drug helps to suppress the inflammatory lesions.

In RA, steroids are very effective, but the recognition of a defective pituitary–adrenal feedback loop in these patients has inspired a novel approach aiming to restore normal corticosteroid levels by a depot of methylprednisolone (Depomedrone) which delivers a low daily dose—the earlier in the disease the better. This treatment accelerates the induction of remission and decreases the side-effects of second-line agents such as gold salts. Selectins and adhesion molecules on endothelial cells and leukocyte integrins appear to be downregulated and this would seriously impede the influx of inflammatory cells into the joint. Anti-inflammatory drugs such as salicylates, innumerable synthetic prostaglandin inhibitors and metalloproteinase poisons are widely used. The so-called second-line drugs, sulfasalazine, penicillamine, gold salts and antimalarials such as chloroquine, all find an important place in therapy but their mode of action is unknown.

Treatment with antibodies to adhesion molecules such as CD44 effectively blocks experimental arthritis

but there are considerable practical and economic problems in adapting to human disease. Therapeutic blocking of other mediators directly concerned in immunological tissue damage will be feasible as cytokine and complement antagonists become available. Neutralizing TNF with a humanized monoclonal antibody is most effective in the short term, so revealing the pathogenetic role of this cytokine. Most significantly, synergistic administration with methotrexate does seem to offer more lasting benefit (figure 20.21).

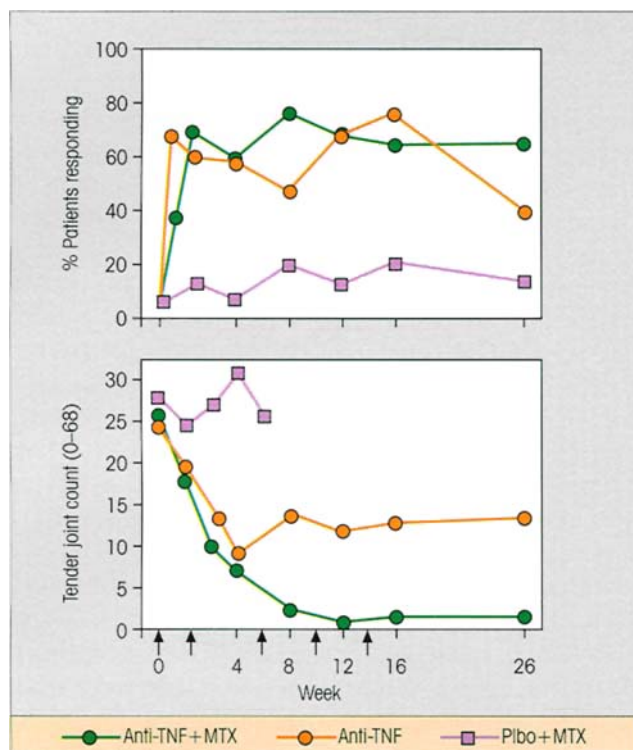


Figure 20.21. Synergy of anti-TNF and methotrexate in the treatment of rheumatoid arthritis. Top panel: Duration of response to therapy as defined by 20% Paulus criteria at three doses of monoclonal chimeric anti-TNF (infliximab) with and without methotrexate (MTX) and placebo (Plbo) plus MTX. Results shown are the proportion (%) of patients responding at weeks 1, 2, 4, 8, 12, 16 and 26. The Paulus response is achieved by 20% improvement in four out of six of the following: tender joint and swollen joint scores, duration of morning stiffness, erythrocyte sedimentation rate and a two grade improvement in the patient's and observer's assessment of disease severity. Lower panel: Serial measurements (median values) of the tender joint count, before (day 0), during (weeks 1–14) and after (weeks 14–26) treatment. Results are included only up to the point at which $\geq 50\%$ of patients remained in the trial (up to week 6 for the placebo plus MTX group). Arrows indicate the timing of infusions of infliximab at weeks 0, 2, 6, 10 and 14. Methotrexate was given weekly and virtually eradicated the production of antibodies to the human chimeric antibody. (Data kindly provided by Professors R.N. Maini, M. Feldmann *et al.*, see Maini R.N. *et al.* (1998) reproduced with permission from Lippincott, Williams & Wilkins, MD, USA, *Arthritis and Rheumatism* 41, 1552.)

Transfection of synovial cells with the natural IL-1 receptor antagonist IL-1Ra may turn out to be a useful long-term strategy.

Immunosuppressive drugs

In a sense, because it blocks cytokine secretion by T-cells, cyclosporin is an anti-inflammatory drug and, since cytokines like IL-2 are also obligatory for lymphocyte proliferation, cyclosporin is also an antimitotic drug. It is of proven efficacy in uveitis, early type I diabetes, nephrotic syndrome and psoriasis and of moderate efficacy in idiopathic thrombocytopenic purpura, SLE, polymyositis, Crohn's disease, primary biliary cirrhosis and myasthenia gravis. In a double-blind randomized control trial, cyclosporin demonstrated significant though not complete disease suppression over 12 months in a group of previously refractory rheumatoid arthritis (RA) patients. Unfortunately, high toxic doses were used but the synergy with rapamycin is a strong indication for a trial of combined therapy. Leflunomide is a promising new agent for treatment of RA. Its active metabolite inhibits *de novo* rUMP synthesis leading to G1 arrest of cycling lymphocytes.

While awaiting more selective therapy, conventional nonspecific antimitotic agents such as azathioprine, cyclophosphamide and methotrexate, usually in combination with steroids, have been used effectively in SLE, RA, chronic active hepatitis and autoimmune hemolytic anemia for example. High-dose i.v. cyclophosphamide plus adrenocorticotrophic hormone (ACTH) or total lymph node irradiation through its effect on the peripheral immune system either slowed or stopped the advance of disease in approximately two-thirds of progressive multiple sclerosis (MS) patients for 1–2 years, a strong indication that the disease is mediated by immune mechanisms. This is further supported by the unfortunate finding that IFN γ exacerbates disease in the majority. Pulsing MS patients with the antileukocyte humanized monoclonal Campath-1H (anti-CD52) produced a brutal and surprisingly persistent reduction in T-cell numbers. Over a 2-year period virtually no new lesions were detected, although in half the patients there was progression of pre-existing lesions. A startling 33% developed Graves' disease (wow!), although this disturbing statistic was not a feature of many other disorders, including RA, where Campath-1H has been used.

It is true that the EAE animal model can be countered by some very diverse manipulations: the antidepressant, roliprim, suppresses cytokine production, a non-hypocalcemic derivative of vitamin D₃ potentially blocks

IL-12 secretion, type IV phosphodiesterase inhibitors suppress Th1 cells and blockade of non-*N*-methyl-D-aspartate (NMDA) receptors ameliorates the neurological sequelae. The problem is that, as it is so difficult to carry out clinical trials in MS, few if any of these approaches are likely to see the light of day.

Immunological control strategies

Cellular manipulation

It should one day be practical to correct any relevant defects in stem cells or in thymus processing by gene therapy, bone marrow or thymus grafting or perhaps, in the latter case, by thymic hormones. Many centers are trying out autologous stem cell transplantation following hemato-immunoablation by cytotoxic drugs in severe cases of autoimmune disease. Around two-thirds of a series of difficult cases of SLE, scleroderma, juvenile and adult RA and so on, stabilized or improved. Transplant-related mortality risk at 2 years was $8 \pm 6\%$, comparable to that seen with cancer patients.

If defects in programmed cell death in antigen-activated T-cells contribute in any way to the development of certain autoimmune diseases, bisindolylmaleimide, which potentiates weak and moderate apoptotic signals, might be therapeutic.

Because T-cell signaling is so pivotal, it is the target

for many strategies (figure 20.22). Injection of monoclonal anti-MHC class II and anti-CD4 successfully fend off lupus in spontaneous mouse models, and it is relevant to record the preliminary clinical observations that injection of immunoglobulins eluted from placentas, and shown to contain anti-allo-class II, significantly ameliorates the symptoms of RA. Immunization with a cyclic peptide from a polymorphic region of the β chain of the Nonobese diabetic (NOD) class II molecule protected a large proportion of the mice from disease. It is unclear whether the mechanism involves a block on MHC antigen presentation or relates to the MHC mimicry hypothesis (cf. p. 413).

Some take the anti-IL-2 receptor approach to deplete activated T-cells (figure 20.22 (1)), but we would like to refer back to our discussion of the long-lasting effect of *nondepleting* anti-CD4 for the induction of tolerance (figure 20.22 (2) and (3)), particularly when reinforced by repeated exposure to antigen (cf. p. 361). Antigen reinforcement of course is an obvious continuing feature in autoimmune disease, so that anti-CD4 should be ideal as a therapy in disorders where the natural 'switch-off' tolerogenic signals are still accepted by the CD4 cells. Ongoing trials in RA still look promising. Excellent remissions have been recorded in patients with Wegener's granulomatosis who were refractory to normal treatment, following sequential injection of anti-CD52 (Campath-1H) and nondepleting anti-CD4 monoclonals.

We can manage perfectly well in life without a complete set of our TCR $V\beta$ genes; after all, wild mice, and presumably ourselves, delete large tracts of $V\beta$ families during thymic differentiation and it does not seem to do them much harm. So, the argument runs, if the autoimmune T-cell clones specific for the autoantigen in a given disease happen to be restricted to membership of a particular $V\beta$ family, we could delete all members of that family *in vivo* with the appropriate antiserum and yet not make irreparable holes in the host's defenses (figure 20.22 (6)). In PL mice immunized with the N-terminal peptide of myelin basic protein, anti- $V\beta 8$ eliminated experimental autoallergic encephalitis almost entirely, so the strategy can work. Not so good with SJL mice which respond to peptide 89–101 with 50% of the T-cells using $V\beta 17$ receptors; in this case, anti- $V\beta 17$ did not block disease. Clearly, until we have a wider knowledge of the extent of $V\beta$ restriction for each antigen and also for each individual, we must suspend judgment on the general feasibility of the strategy.

Now, if one takes the view that rheumatoid factor immune complexes are major players in the patho-

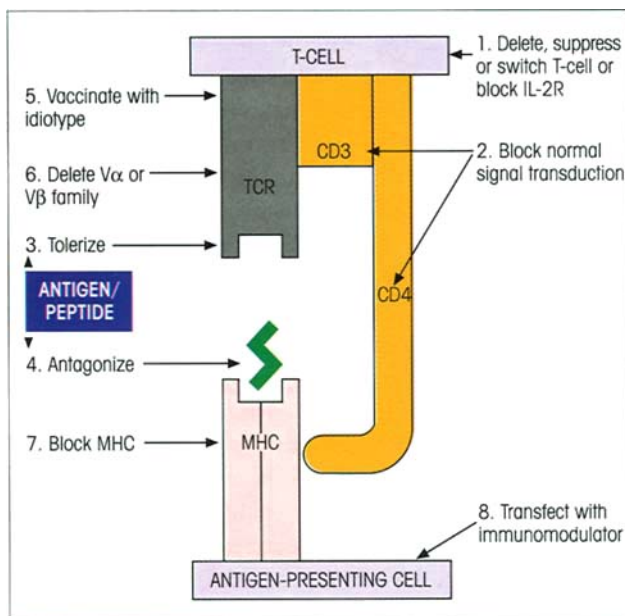


Figure 20.22. Strategic options for therapy based on T-cell targeting.